



Comparisons of two-phase condensation and single-phase heat transfer and frictional pressure drop characteristics and energy-saving performance analysis of R-32 and R-410A in plate heat exchanger

Joo Young Song^a, Dong Ho Kim^b, Chan Ho Song^b, Yong Tae Kang^{c,*}

^a Graduate School of Mechanical Engineering, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea

^b Energy Systems Research Division, Korea Institute of Machinery and Materials, 156 Gajeongbuk-ro, Yuseong-gu, Daejeon, 34103, Republic of Korea

^c Department of Mechanical Engineering, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea

ARTICLE INFO

Keywords:

Two-phase condensation
Single-phase cooling
Plate heat exchanger
Heat transfer coefficient
Frictional pressure drop
Energy-saving performance

ABSTRACT

In this study, the heat transfer and frictional pressure drop characteristics of R-32 and R-410A in a plate heat exchanger are experimentally evaluated. The study comprises two-phase condensation, single-phase vapor, and liquid cooling experiments of R-32 and R-410A, as well as an independent single-phase water heat transfer experiment. The effects of heat flux, mass flux, condensing pressure, mean vapor quality, and refrigerant Reynolds number on heat transfer and pressure drop are assessed for comparisons of R32 and R410A. The results demonstrate that the heat transfer performance and frictional pressure drop increase with rising mass flux and mean vapor quality and decrease with declining condensing pressure in the two-phase condensation experiments. Furthermore, increasing the heat flux enhances the heat transfer performance while having negligible impact on the frictional pressure drop. In the single-phase cooling experiments, both heat transfer performance and frictional pressure drop rise with increasing refrigerant flow rate. Based on the experimental results, Nusselt number and friction factor correlations for two- and single-phase flows of R-32 and R-410A are developed and compared with the correlations from other studies. Finally, energy-saving performance is evaluated and compared between the refrigerants.

1. Introduction

In response to concerns over ozone layer depletion and global warming, the international community is enforcing stricter regulations on refrigerant substitutions. The initial refrigerant regulations, primarily aimed at preventing ozone layer depletion, were enacted through the Montreal Protocol [1] in 1987. However, current refrigerant regulations predominantly focus on mitigating global warming through the F-gas regulation [2], the Kigali amendment [3], the Paris Agreement [4], and others.

R-410A, comprising 50% wt. R-32 and 50% wt. R-125, has a GWP of 2088 [5]. R-452B (GWP 698 [5]) and R-454B (GWP 466 [5]) are HFC/HFO blends with operating pressure ranges similar to R-410A, making them suitable for drop-in replacements for R-410A systems. Guilherme et al. [6] reported increased COP when R-452B and R-454B were introduced into the R-410A system. However, both refrigerants are mildly flammable and possess the ASHRAE safety classification A2L.

R-466A, an HFC/CF3I blend with a GWP of 733 [7], is a non-flammable refrigerant classified under the ASHRAE safety classification A1, making it a promising R-410A substitute. Nonetheless, Guilherme et al. [6] noted that R-466A has a larger molar mass than R-410A, which increases the refrigerant charge in the system, causing energy loss due to pressure drop and increased costs associated with the expansion valves.

Kim et al. [8] compared the R-410A heat pump system with the low-GWP refrigerant heat pump systems through simulation. According to the drop-in simulation results by Kim et al. [8], the system COP increased when R-32 (GWP 675 [9]) and L-41a (GWP 490 [6]) were applied compared to R-410A. Both R-32 and L-41a are classified as mildly flammable, with an ASHRAE safety classification of A2L. R-32 is a pure HFC, while L-41a is an HFO/HFC blend.

Low-GWP refrigerant candidates that can replace R-410A have concerns regarding safety issues, such as flammability and toxicity [6]. Even non-flammable refrigerants like R-466A can pose cost issues due to their properties. In the absence of a clear alternative, R-410A remains one of the most widely used refrigerants internationally [6,10,11].

* Corresponding author.

E-mail address: ytkang@korea.ac.kr (Y.T. Kang).

<https://doi.org/10.1016/j.ijheatmasstransfer.2024.125813>

Received 6 March 2024; Received in revised form 21 May 2024; Accepted 1 June 2024

Available online 8 June 2024

0017-9310/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclatures		Greeks	
A	Area, m^2	β	Chevron angle, $^\circ$
B	Biased uncertainty, %	ΔP	Pressure difference, kPa
b	Corrugation depth, m	ΔT_{LMTD}	Logarithmic mean temperature difference
Co	Condensation number	λ	Corrugation pitch, m
Cp	Specific heat, $kJ/kg \cdot K$	μ	Viscosity, $kg/m \cdot s$
D	Diameter, m	ρ	Density, kg/m^3
EPP	Energy performance parameter	σ	Standard deviation
F	Calculated factor	$\phi(X)$	Enlargement factor
f	Friction factor	X	Dimensionless corrugation parameter
G	Mass flux, $kg/m^2 \cdot s$	ω	Calculated uncertainty, %
g	Gravitational acceleration, m/s^2	Subscripts	
h	Heat transfer coefficient, $kW/m^2 \cdot K$	avg	Averaged
i	Enthalpy, $kJ/kg \cdot K$	c	Cold side
k	Thermal conductivity, $kW/m \cdot K$	ch	Single channel
L	Length, m	$cross$	Cross sectional
$\dot{m}$	Mass flow rate, kg/s	de	Deceleration
N	Number	eff	Effective
Nu	Nusselt number	f	Frictional
P	Pressure, kPa	fg	Difference between liquid and vapor states
Pr	Prandtl number	g	Gravitational
$\dot{Q}$	Heat transfer rate, kW	h	Hot side
R	Random uncertainty, %	$height$	Height
Re	Reynolds number	hyd	Hydraulic
T	Temperature, $^\circ C$	i	Inlet
t	Plate thickness, m	l	Liquid state
U	Measurement data uncertainty, %	m	Mean value
$U_{overall}$	Overall heat transfer coefficient, $kW/m^2 \cdot K$	$measured$	Measured value
V	Flow velocity, m/s	o	Outlet
v	Specific volume, m^3/kg	p	Single plate
$\bar{X}$	Mean measured value	$port$	Manifold port
X_n	n^{th} measured datum in a dataset	pre	Pre-condenser
x	Vapor quality	$proj$	Projected
Acronyms		r	Refrigerant
CP	Condensing pressure	sp	Single-phase
FPD	Frictional pressure drop	$test$	Test section
HF	Heat flux	tot	Total
HTC	Heat transfer coefficient	tp	Two-phase
MF	Mass flux	v	Vapor state
VQ	Vapor quality	w	Water
PHE	Plate heat exchanger	$wall$	Wall
SP	Single-phase	$width$	Width
TP	Two-phase		

While it is important to reduce the use of high-GWP refrigerants in accordance with environmental regulations, it is also crucial to enhance the efficiency of those currently in use.

The plate heat exchanger (PHE), developed in the 1920s, was initially employed for moderate temperature, low pressure, single-phase (SP) heat transfer due to limitations in gasket materials. However, Copper-brazed PHEs are suitable for heat transfer in higher temperature and pressure ranges compared to gasketed PHEs. Through continuous development, modern PHEs offer chemical resistance by using titanium or stainless steel as the bonding material instead of copper.

Longo and Gasparella [12], Hsieh and Lin [13], Yan and Lin [14], and Kwon et al. [15] developed SP water heat transfer correlations to evaluate two-phase (TP) refrigerant characteristics in PHEs. Khan et al. [16] and Yang et al. [17] investigated the effect of chevron angle on heat transfer of water in PHEs. Rao et al. [18] and Muley and Manglik [19] evaluated the thermal and hydraulic performance of water in PHEs.

Nilpueng and Wongwises [20] studied the impact of heat transfer plate roughness on thermal and hydraulic characteristics in a PHE.

Moreover, some studies evaluated liquid flow characteristics of fluids other than water in PHEs. Bhattad et al. [21] and Teng et al. [22] assessed nanofluids' thermal characteristics, while Song et al. [23,24] examined aqueous lithium-bromide characteristics and published correlations. Hsieh and Lin [25] evaluated the thermal performance of subcooled liquid R-410A and developed a heat transfer coefficient (HTC) correlation.

Furthermore, there have been several studies focusing on heat transfer properties of fluids on various surfaces. Khoshvaght-Aliabadi et al. [26-30] conducted a numerical evaluation of the thermal performance of carbon dioxide in various plate minichannels. Park et al. [31, 32] carried out experimental assessments of the pool boiling characteristics of HFE-7100 on both hydrophilic and hydrophobic surfaces. Kim et al. [33] investigated the boiling performance of HFE-7000 on

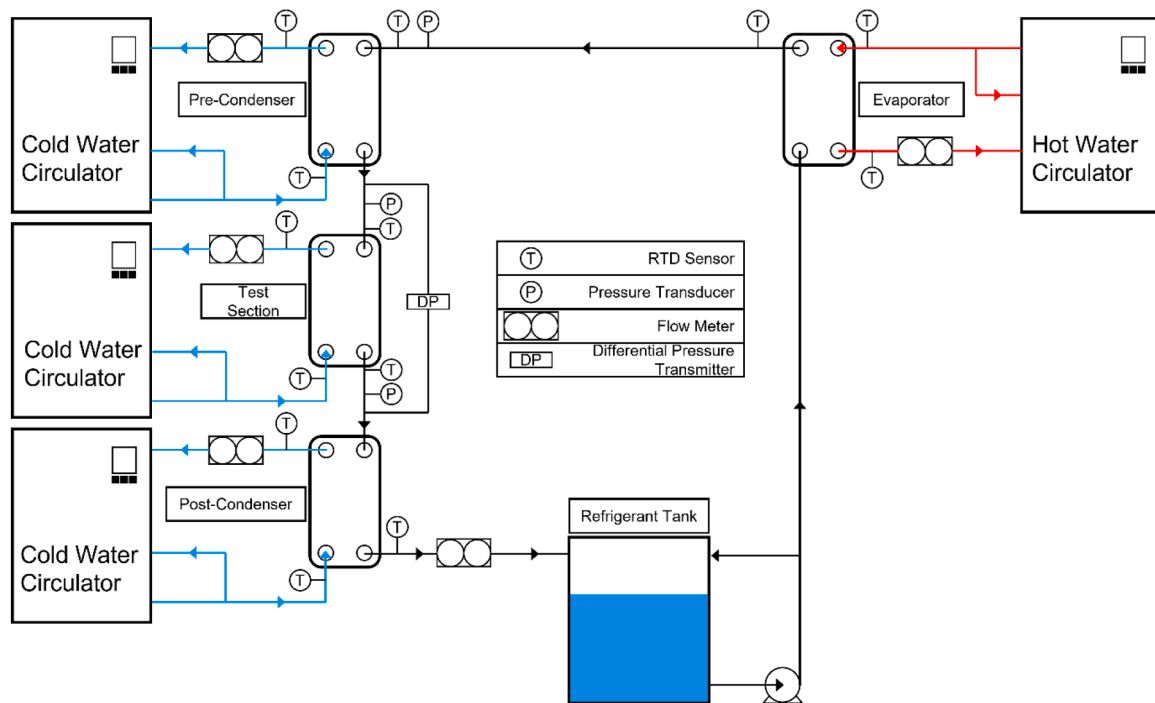


Fig. 1. Schematic diagram of refrigerant cooling experimental setup.

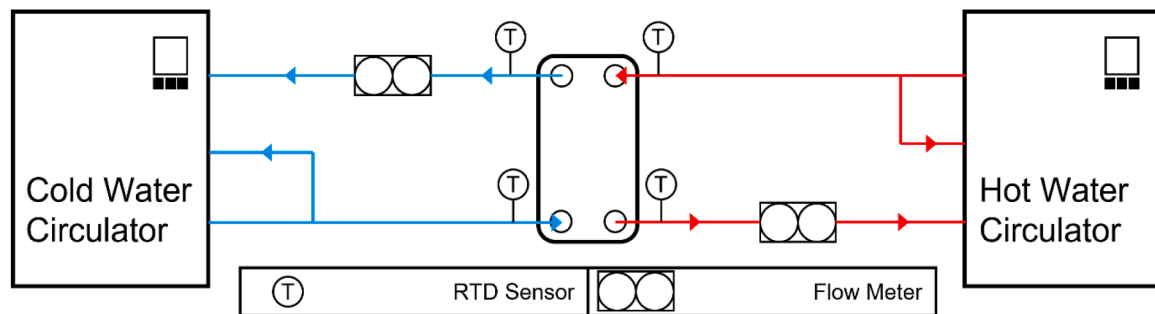


Fig. 2. Schematic diagram of water heat transfer experimental setup.

nano-textured surfaces. Ko et al. [34] conducted an comprehensive literature review on the thermal and hydraulic performance of air on various shapes of fin surfaces.

Hsieh and Lin [25], Han et al. [35], and Kuo et al. [36] developed correlations predicting TP R-410A HTC and frictional pressure drop (FPD). Longo [37] compared the TP condensation performance of R-410A [38] with those of R-236fa and R-134a. Subsequently, Longo et al. [39,40] compared the TP thermal performances of R-410A with those of R-32.

In addition to studies on R-410A, there are also investigations evaluating TP characteristics in PHEs for other refrigerants. Longo and his colleagues [12,41,42] assessed the TP characteristics of R-134a and R-1234ze(Z) in a PHE. The studies [12,37–42] utilized the same PHE in their research.

Vlasogiannis et al. [43] visualized the TP flow in a PHE and discussed the shear-effect of the fluid according to the flow rate. Jung et al. [44] evaluated R-1234ze (E) condensation in the same PHE as Kwon et al. [15], and their results were compared to those of R-1233zd (E) in Kwon et al. [15]. The two studies [15,44] addressed the importance of the shear-effect on the TP flow. In addition, there are two studies [45,46] utilized the R-1233zd(E) condensation experimental results from Kwon et al. [15].

The actual condensation process encompasses the vapor state, TP

state, and liquid state. However, only one study has published the SP R-410A characteristics in a PHE, and it is limited to liquid state cooling of hot water. In this study, the TP condensation thermal and hydraulic characteristics of R-32 and R-410A, as well as those of each liquid and vapor state, were evaluated. Moreover, the economics in terms of energy-saving was analyzed. The findings of this study were compared with other studies and validated its energy-saving aspect.

2. Experiment

2.1. Experimental apparatus

The refrigerant experimental setup of this study was built, and the corresponding schematic diagram is presented in Fig. 1. The refrigerant flows in the following order: refrigerant tank, evaporator, pre-condenser, test PHE, and post-condenser. The refrigerant, initially in liquid within the refrigerant tank, circulates through the magnetic gear pump. After passing through the pump, the liquid refrigerant enters the evaporator, where it is completely evaporated via heat exchange with hot water. The hot water is heated by six 1.5 kW cartridge heaters installed in the hot water circulator. The superheated vapor refrigerant, fully evaporated in the evaporator, enters the pre-condenser, where it is partially condensed by heat exchange with cooling water. The vapor

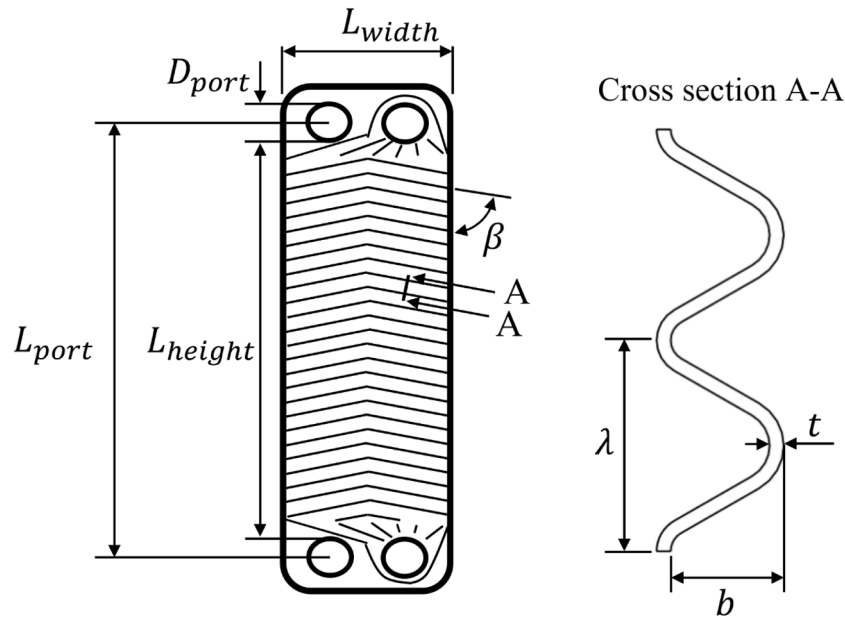


Fig. 3. Schematic diagram of test PHE.

Table 1
Geometric conditions of the test PHE.

Item	Nomenclature	Value
Number of plates	N_p	10
Port to port length, m	L_{port}	0.243
Plate height, m	L_{height}	0.2185
Plate width, m	L_{width}	0.117
Port diameter, m	D_{port}	0.0245
Corrugation pitch, m	λ	0.0075
Corrugation depth, m	b	0.00197
Plate thickness, m	t	0.0005
Number of channels of refrigerant side	$N_{ch,r}$	4
Number of channels of water side	$N_{ch,w}$	5
Chevron angle, °	β	65

Table 2
Summary of measurements and maximum uncertainties.

Items	Specifications and uncertainties
Temperature	Model: PT100Ω RTD Accuracy: $\pm 0.15 + 0.002T$ Range: $-30 \sim 300$ °C
Volumetric flow rate	Model: FLUID-EX oval gear flowmeter Accuracy: ± 0.5 % of full scale Range: $0 \sim 20$ LPM
Pressure	Model: PHP pressure sensor Accuracy: ± 0.035 % of full scale Range: $1 \sim 3.5$ bar
Differential pressure	Model: DNS-020 Accuracy: ± 0.015 % of full scale Range: $0 \sim 35$ kPa
ω_{Nu}	9.55 % (R-410A) 8.43 % (R-32)
ω_{HTC}	8.60 % (R-410A) 7.37 % (R-32)
ω_f	5.83 % (R-410A) 5.87 % (R-32)
ω_{FPD}	3.08 (R-410A) 3.51 % (R-32)

quality (VQ) of the TP refrigerant is adjusted through partial condensation in the pre-condenser. The TP refrigerant then undergoes condensation via heat exchange with cooling water in the test section.

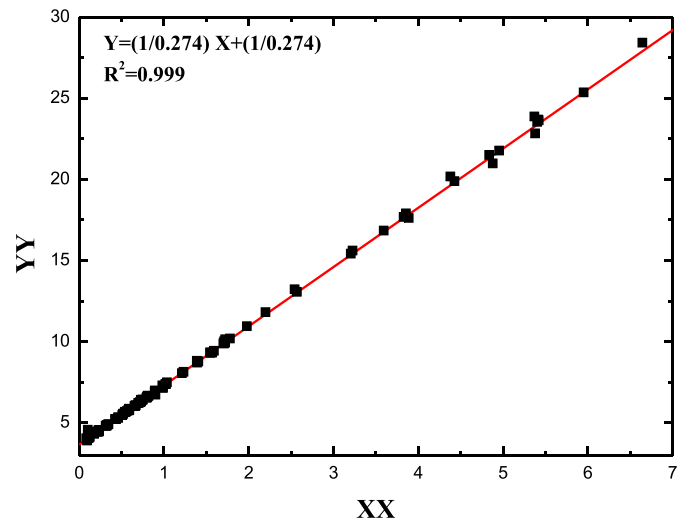


Fig. 4. Modified Wilson plot result of water heat transfer data.

Temperature, pressure, and pressure difference at the inlet and outlet are measured to evaluate the condensation thermal and hydraulic characteristics of the refrigerant. The TP refrigerant subsequently becomes fully condensed into the liquid state through heat exchange with cooling water in the post-condenser. The volumetric flow rate of the fully condensed liquid refrigerant is measured by an oval gear volumetric flow meter before returning to the refrigerant tank. All heat exchangers are plate heat exchangers installed vertically from the ground. Water flow rates are recorded at the outlets of the heat exchangers.

The liquid and vapor state SP refrigerant cooling thermal and hydraulic performances are also evaluated in the experimental setup in Fig. 1. For the liquid refrigerant cooling process, the liquid refrigerant is heated and pressurized in the evaporator. The fully vaporized refrigerant subsequently becomes fully condensed to subcooled liquid in the pre-condenser. The subcooled and temperature-controlled liquid refrigerant then enters the test section. In the vapor phase refrigerant cooling experiment, the liquid refrigerant completely evaporates in the evaporator, and the resulting superheated vapor refrigerant enters the pre-condenser without transferring heat with the cooling water. The

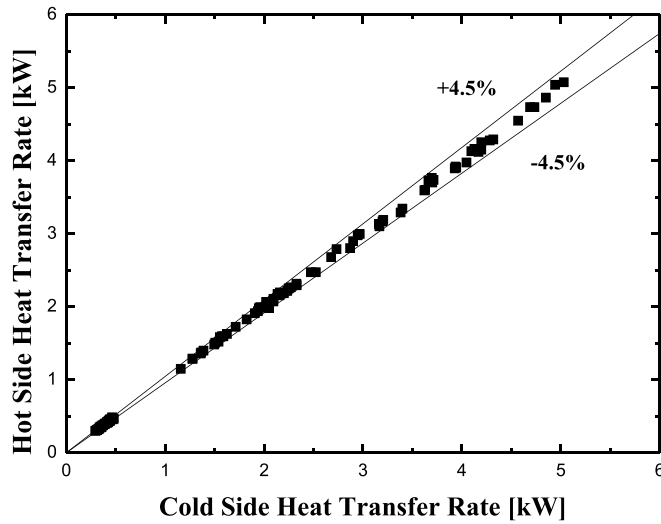


Fig. 5. Energy balance error in water heat transfer experiment.

Table 3

SP water heat transfer experiment conditions.

	Cold Water	Hot Water
Inlet Temp., °C	17.3 - 31.3	30.8 - 48.1
Mass Flow Rate, g/s	4.5 - 122.8	5.5 - 142.8
Reynolds Number, -	15.3 - 509.5	24.5 - 922.4
Prandtl Number, -	4.3 - 7.4	3.7 - 6.2

uncooled superheated vapor refrigerant is then cooled by the cooling water in the test section before condensing into a TP refrigerant.

Fig. 2 presents a schematic diagram of the water heat transfer experimental setup used in this study. Hot and cold waters flow through the same test PHE as in Fig. 1. In the same way that the cooling water in Fig. 1, the cold water in Fig. 2 also flows upward in the test section. The hot water flows downward in the test section, opposite to the cold water. The temperatures at the inlet and outlet of the test section are measured for both hot and cold waters, and the flow rates are measured at each outlet through oval gear volumetric flow meters.

Fig. 3 presents a schematic diagram of the test PHE. The internal geometric parameters of the test PHE are summarized in Table 1. The test PHE comprises 10 plates in total, of which 8 participate in heat exchange, excluding 2 end plates. The ports of the test PHE consist of 5 and 4 channels, respectively. The refrigerant and cooling water are arranged in counterflow. The refrigerant flows downward through the 4 channel ports, and the cooling water flows upward through the 5 channel ports in the test PHE.

2.2. Data reduction

2.2.1. Test phe

The enlargement factor $\phi(X)$ is calculated as Eq. (1) as [47].

$$\phi(X) \approx \frac{1}{6} \left(1 + \sqrt{1 + X^2} + 4\sqrt{1 + \frac{X^2}{2}} \right) \quad (1)$$

The dimensionless corrugation parameter X in Eq. (1) is calculated as Eq. (2).

$$X = \frac{\pi b}{\lambda} \quad (2)$$

The hydraulic diameter D_{hyd} is calculated as Eq. (3).

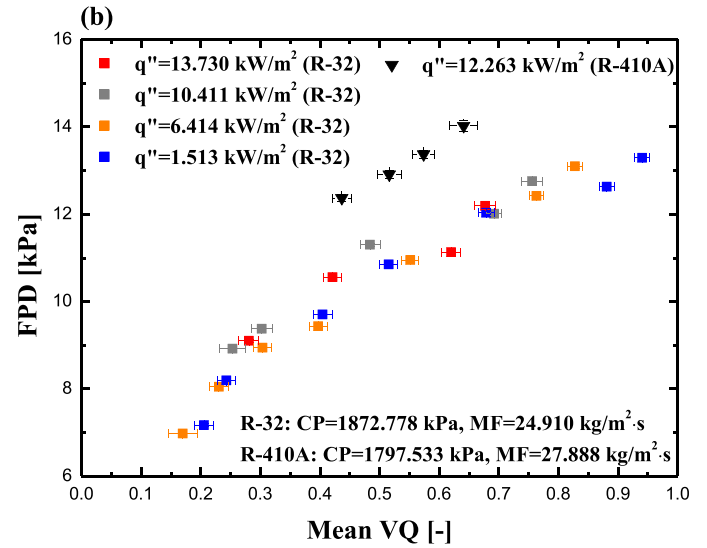
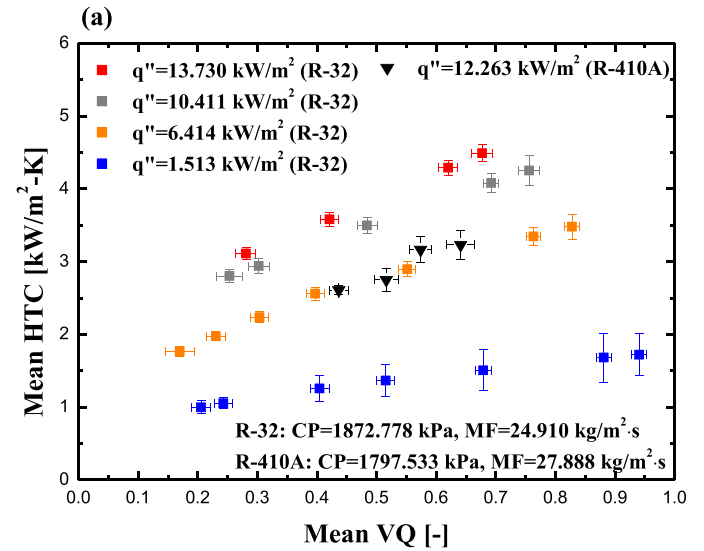


Fig. 6. Effects of HF and mean VQ on (a) mean HTC and (b) FPD.

Table 4

R-32 TP condensation experimental conditions.

	Refrigerant side	Cooling water side
Inlet temperature, °C	28.807–46.705	14.181–37.024
Condensing pressure, kPa	1771.938–2750.67	–
Pressure difference, kPa	0.939–10.908	–
Mass flow rate, g/s	16.46–29.21	7.494–131.4
Mean vapor quality, -	0.063–0.965	–
Reynolds number, -	732.1–3797 (Equivalent)	26.09–575.5
Prandtl number, -	1.855–1.988 (Liquid)	4.289–7.135

$$D_{hyd} = \frac{2b}{\phi(X)} \quad (3)$$

The total heat transfer area A_{tot} is calculated as Eq. (4).

$$A_{tot} = (N_p - 2) \times A_{eff} \quad (4)$$

The effective heat transfer area of a single plate A_{eff} in Eq. (4) is

Table 5
R-410A TP condensation experimental conditions.

	Refrigerant side	Cooling water side
Inlet temperature, °C	28.570–46.539	19.577–37.229
Condensing pressure, kPa	1745.26–2708.93	–
Pressure difference, kPa	3.147–16.897	–
Mass flow rate, g/s	18.35–29.82	4.479–123.7
Mean vapor quality, -	0.047–0.946	–
Reynolds number, -	694.2–3160 (Equivalent)	15.52–425
	592.2–1107 (Liquid)	
Prandtl number, -	2.378–2.609 (Liquid)	4.219–6.568

calculated as Eq. (5).

$$A_{eff} = \phi(X) \times A_{proj} \quad (5)$$

The projected heat transfer area of a single plate A_{proj} in Eq. (5) is calculated as Eq. (6).

$$A_{proj} = L_{height} \times L_{width} \quad (6)$$

The cross-sectional area of the manifold ports A_{port} is calculated as

Eq. (7).

$$A_{port} = \frac{\pi \cdot D_{port}^2}{4} \quad (7)$$

The cross-sectional channel area A_{cross} is calculated as Eq. (8).

$$A_{cross} = b \cdot L_{width} \quad (8)$$

2.2.2. Heat transfer data reduction

The average heat transfer rate is expressed as shown in Eq. (9) below.

$$\dot{Q}_{avg} = U_{overall} \cdot A_{tot} \cdot \Delta T_{LMTD} \quad (9)$$

ΔT_{LMTD} in Eq. (1) is calculated as shown in Eq. (10) below.

$$\Delta T_{LMTD} = \frac{(T_{h,i} - T_{c,o}) - (T_{h,o} - T_{c,i})}{\ln\left(\frac{T_{h,i} - T_{c,o}}{T_{h,o} - T_{c,i}}\right)} \quad (10)$$

The overall HTC $U_{overall}$ in Eq. (9) can be expressed as shown in Eq. (11) below [48].

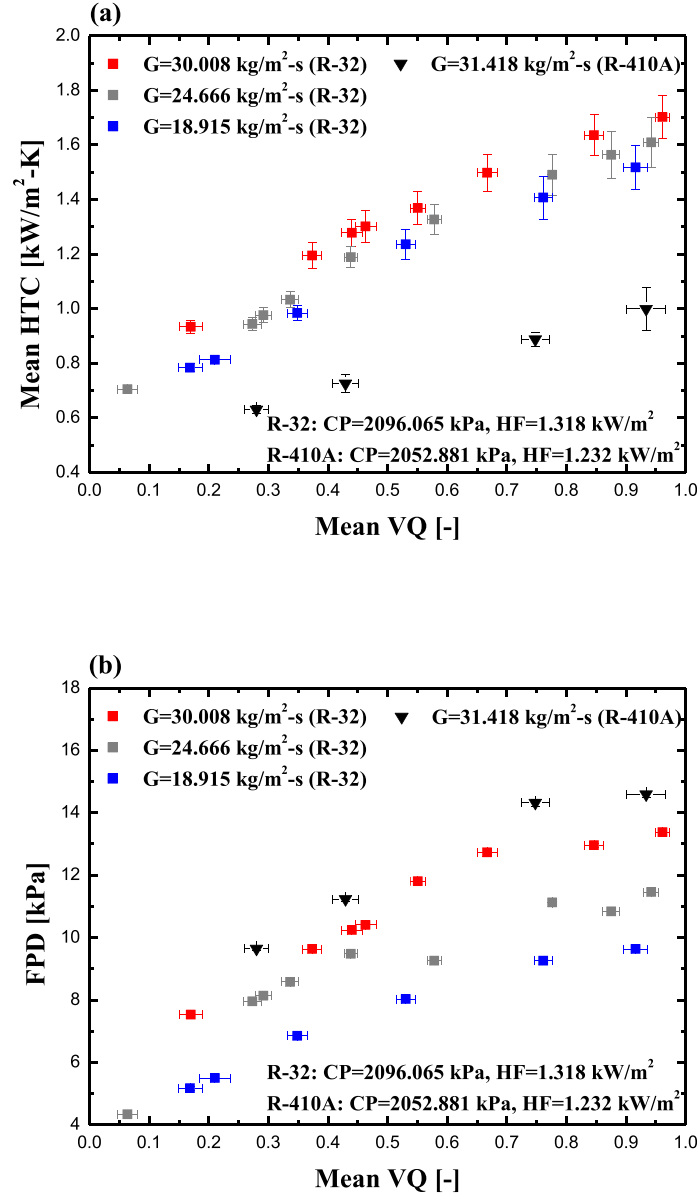


Fig. 7. Effects of MF and mean VQ on (a) mean HTC and (b) FPD.

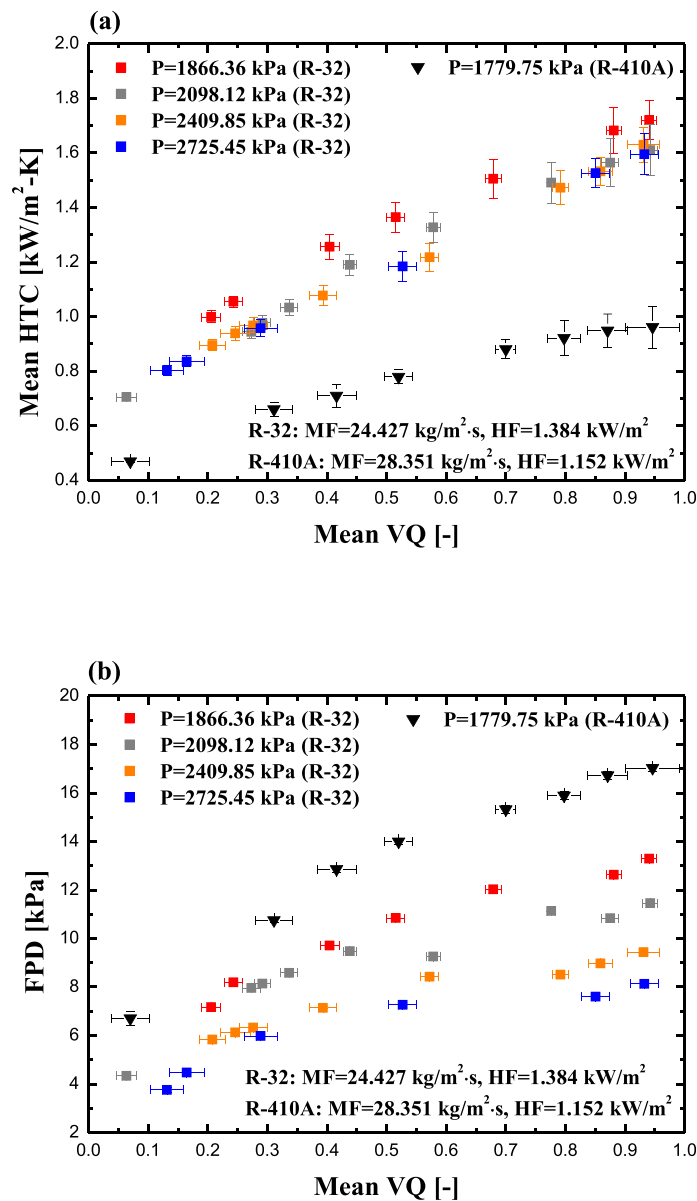


Fig. 8. Effects of CP and mean VQ on (a) mean HTC and (b) FPD.

Table 6
R-32 & R-410A thermal properties with various saturation pressures.

Refrigerant			R-410A	R-32	R-32	R-32	R-32
Condensing pressure		kPa	1779.75	1866.36	2098.12	2409.85	2725.45
Saturation temperature		°C	7.82	28.76	33.31	38.86	43.95
Latent heat		kJ/kg	182.1	263.1	253.1	239.9	226.8
Specific heat capacity	Liq.	kJ/kg·K	1.736	1.981	2.044	2.140	2.253
	Vap.	kJ/kg·K	1.502	1.683	1.794	1.961	2.158
Thermal conductivity	Liq.	kW/m·K·10 ⁶	86.17	119.6	115.4	110.2	105.2
	Vap.	kW/m·K·10 ⁶	15.96	18.11	18.97	20.13	21.34
Viscosity	Liq.	kg/m·s·10 ⁶	118.2	112.5	106.4	99.16	92.63
	Vap.	kg/m·s·10 ⁶	13.36	12.98	13.25	13.63	14.01
Density	Liq.	kg/m ³	1046	945	924.8	898.7	872.9
	Vap.	kg/m ³	71.72	52.84	60.3	70.86	82.25

Table 7

SP R-32 cooling experimental conditions.

		Refrigerant side	Cooling water side
Vapor state	Inlet temperature, °C	45.348–46.844	40.636–42.578
	Pressure difference, kPa	1.165–2.154	–
	Mass flow rate, g/s	11.190–21.190	7.928–7.992
	Reynolds number, -	2935–5588	37.65–38.51
	Prandtl number, -	1.414–1.446	4.025–4.148
Liquid state	Inlet temperature, °C	21.307–31.851	8.865–21.176
	Pressure difference, kPa	0.008–1.607	–
	Mass flow rate, g/s	12.77–28.61	7.99–40.98
	Reynolds number, -	364.9–911.9	20.64–125.4
	Prandtl number, -	1.841–1.861	6.224–9.497

Table 8

SP R-410A cooling experimental conditions.

		Refrigerant side	Cooling water side
Vapor state	Inlet temperature, °C	45.225–45.426	40.763–41.068
	Pressure difference, kPa	1.536–2.509	–
	Mass flow rate, g/s	14.91–25.85	4.437–4.63
	Reynolds number, -	3556–6136	21.13–22.17
	Prandtl number, -	1.678–1.696	4.088–4.112
Liquid state	Inlet temperature, °C	22.455–26.760	19.625–20.727
	Pressure difference, kPa	0.012–2.426	–
	Mass flow rate, g/s	11.22–31.38	4.822–40.17
	Reynolds number, -	346.1–975.4	15.39–120.8
	Prandtl number, -	2.374–2.384	6.212–6.942

$$\frac{1}{U_{overall}} = \frac{1}{h_h} + \frac{1}{h_c} + \frac{t}{k_{wall}} \quad (11)$$

The general form of the SP Nu correlation can be expressed as shown in Eq. (12) [49].

$$Nu_{sp} = C_1 \cdot Re^{C_2} \cdot Pr^{0.3333} \quad (12)$$

The definitions of SP Re and Pr in Eq. (12) are shown in Eqs. (13) and (14).

$$Re = \frac{\rho \cdot V \cdot D_{hyd}}{\mu} = \frac{G_{ch} \cdot D_{hyd}}{\mu} \quad (13)$$

$$Pr = \frac{\mu \cdot Cp}{k} \quad (14)$$

For simple and practical data reduction and analysis, this study assumed that all TP flows were homogeneously mixed. Consequently, all equations related to TP flow are calculated using the mean vapor quality. The inlet vapor quality was controlled by a pre-condenser and the outlet vapor quality was determined from the inlet vapor quality and the heat flux from the cold water loop. The filmwise condensation HTC correlation for a plane wall was firstly established by Nusselt [50]. Visualization results indicate that TP downward flow condensation predominantly occurs in a filmwise manner [51–53]. Assuming the film condensation in a plate heat exchanger, it is practical to analyze the effects of various mean vapor qualities on the HTC and FPD. Consequently, many studies analyze TP condensation in plate heat exchangers as filmwise [36,38,41,54]. In this study, downward TP flow condensation was considered filmwise, and the results were analyzed accordingly.

The form of the TP Nu correlation developed in this study is as shown in Eq. (15).

$$Nu_{tp} = C_1 \cdot Re_{eq}^{C_2} \cdot Co^{C_3} \cdot Pr_l^{0.3333} \quad (15)$$

The liquid state Pr_l is calculated as Eq. (14). The definitions of Re_{eq} and Co are as shown in Eqs. (16)–(19).

$$Re_{eq} = \frac{G_{eq} \cdot D_{hyd}}{\mu_l} \quad (16)$$

$$G_{eq} = G_{ch} \cdot \left(1 - x_m + x_m \cdot \left(\frac{\rho_l}{\rho_v} \right)^{0.5} \right) \quad (17)$$

$$G_{ch} = \rho \cdot V = \frac{\dot{m}}{N_{ch} \cdot A_{cross}} \quad (18)$$

$$Co = \frac{(i_i - i_o)}{\Delta i_{fg}} \quad (19)$$

2.2.3. Frictional pressure drop data reduction

The measured total pressure drop consists of FPD ΔP_f , the pressure drop in the ports ΔP_{port} , the deceleration pressure drop due to condensation phase change ΔP_{de} , and the gravitational pressure drop from downward flow ΔP_g . It can be expressed as shown in Eq. (20).

$$\Delta P_{measured} = \Delta P_f + \Delta P_{port} - \Delta P_{de} - \Delta P_g \quad (20)$$

ΔP_{port} in Eq. (20) is calculated as Eq. (21) according to Kays and London [55] and Shah [56].

$$\Delta P_{port} = 1.5 \cdot \frac{u_m^2}{2 \cdot v_m} \quad (21)$$

ΔP_{de} is calculated as shown in Eq. (22).

$$\Delta P_{de} = G_{ch}^2 \cdot v_{fg} \cdot \Delta x \quad (22)$$

In SP flow, since there is no phase change, ΔP_{de} in Eq. (22) is neglected.

ΔP_g of the TP flow is calculated as Eq. (23).

$$\Delta P_g = \frac{g \cdot L_{height}}{v_{r,m}} \quad (23)$$

The form of the SP f correlation is expressed as shown in Eq. (24).

$$f_{sp} = C_1 \cdot Re^{C_2} \quad (24)$$

The form of the TP f correlation is expressed as shown in Eq. (25).

$$f_{tp} = C_1 \cdot Re_{eq}^{C_2} \cdot Re_l^{C_3} \quad (25)$$

SP and liquid state Re in Eqs. (24) and (25) is calculated as Eq. (13).

2.2.4. Uncertainty analysis

The uncertainty U of the measured values is calculated using the systematic uncertainty B and the random uncertainty R as shown in Eq. (26) below [57].

$$U = \sqrt{B^2 + R^2} \quad (26)$$

Systematic uncertainty B is provided in the measuring instrument's manual, and random uncertainty R is calculated using the standard deviation σ and Z-score. The Z-score for a 95 % confidence level is 2, and based on this, R is calculated as shown in Eq. (27).

$$R = \frac{2 \cdot \sigma}{\bar{X}} \quad (27)$$

The uncertainty of the calculated value F is determined using the impact of the parameter x_i on F and the uncertainty of x_i as shown in Eq. (28).

$$\omega_F = \sqrt{\sum \left(\frac{\partial F}{\partial x_i} \omega_{x_i} \right)^2} \quad (28)$$

To evaluate the thermal and hydraulic characteristics, all measurements are recorded once every 5 s in a steady state. The specifications of the instruments and the uncertainties of the calculated values are summarized in Table 2.

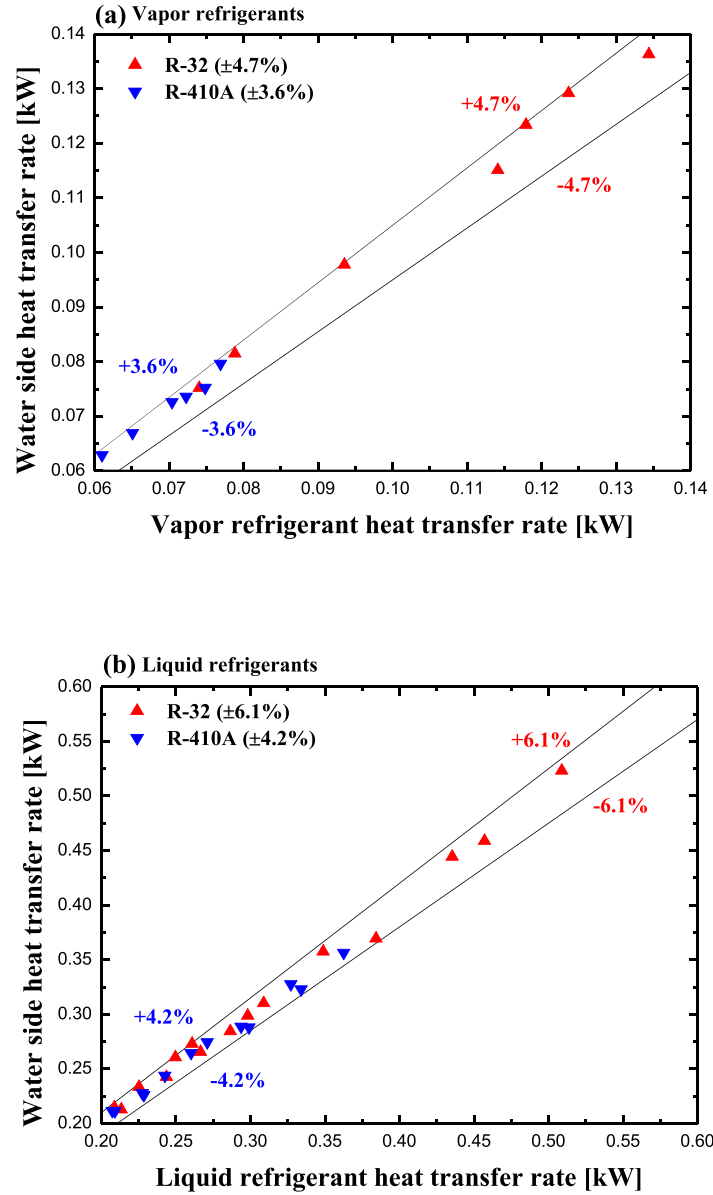


Fig. 9. Energy balance errors of SP (a) vapor and (b) liquid refrigerants.

3. Results and discussion

3.1. Single-phase (SP) water heat transfer

The results of the water heat transfer experiments in this study, when applied to the modified Wilson plot method procedure of Song et al. [23], led to the development of a water Nu correlation for the test-PHE, as Eq. (29).

$$Nu_w = 0.274 \cdot Re_w^{0.687} \cdot Pr_w^{0.333} \quad (29)$$

Eq. (29), Nu_w correlation, is applicable for Reynolds numbers in the range of 15.06 - 922.4 and Prandtl numbers in the range of 3.718 - 7.444 and R-square value was 0.999. A graph applying the modified Wilson plot method to the water heat transfer experiment results is shown in Fig. 4. Water heat transfer experiment was conducted within $\pm 4.5\%$ of energy balance error range and is shown in Fig. 5. Water heat transfer experimental conditions are summarized in Table 3.

3.1. Two-phase (TP) condensation results

Figs. 6 to 8 demonstrate the effects of heat flux (HF), mass flux (MF), and condensing pressure (CP) on two-phase (TP) R-32's (a) mean heat transfer coefficient (HTC) and (b) frictional pressure drop (FPD). The impacts of HF, MF, and CP on the flow properties of condensing TP R-32 were compared with the condensing TP R-410A results.

In the condensation process of the TP refrigerant, shear stress increases with increasing VQ of the refrigerant [15,36,41,44,58]. According to Vlasogiannis et al. [43] and Jiang and Bai [59], as the shear of a TP fluid increases, both the HTC [43] and the FPD [59] increase. These results were also observed in this study. As shown in the graphs in Figs. 6 to 8, the HTC increases as the mean VQ increases, and the FPD follows the same pattern. This is because, as the VQ of the refrigerant increases, the shear applied by the vapor to the liquid on the wall of the test PHE intensifies, and the secondary swirling flow of the liquid film on the test PHE wall correspondingly increases. The experimental conditions for TP R-32 and R-410A condensation are detailed in Tables 4 and 5, respectively.

Fig. 6 illustrates the influence of varying HF on (a) mean HTC and (b)

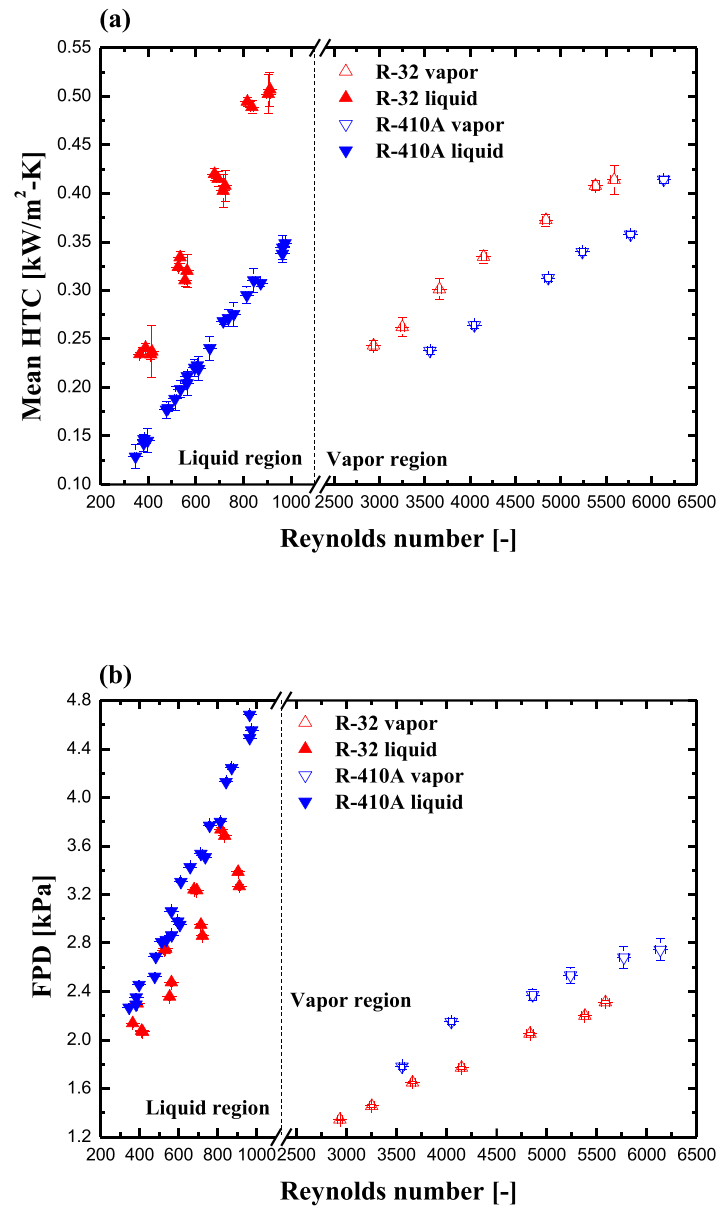


Fig. 10. Vapor R-32 and R-410A cooling (a) mean HTC and (b) FPD with various Reynolds numbers.

FPD. The experiments were conducted within HF ranges of 1.224–25.24 kW/m² for R-32 and 0.672–12.970 kW/m² for R-410A. The figure depicts the mean HTC and FPD at R-32 HF range of 1.513–13.730 kW/m². The average saturation temperature for the results shown is 28.89 °C, with an average CP of 1872.778 kPa and a MF of 24.910 kg/m²•s. Mean HTC and FPD results of R-410A, at a HF of 12.263 kW/m², which were obtained under similar conditions to R-32, are also displayed. In the similar HF conditions, R-32's mean HTC was higher than that of R-410A, and R-32's FPD was lower.

Fig. 7 displays the effects of MF variation on (a) mean HTC and (b) FPD. Experiments were conducted within MF ranges of 17.85–31.69 kg/m²•s for R-32 and 19.91–32.35 kg/m²•s for R-410A. The figure presents the mean HTC and FPD at R-32 MF range of 18.915–30.008 kg/m²•s. The average saturation temperature for these results is 33.27 °C, with an average CP of 2096.065 kPa and an HF of 1.318 kW/m². R-410A results at an MF of 31.418 kg/m²•s, obtained under similar conditions, are also presented. Under similar MF conditions, the mean HTC of R-32 was higher and the FPD was lower than that of R-410A.

Fig. 8 presents the influence of CP variation on (a) mean HTC and (b) FPD. Experiments were conducted within CP ranges of

1771.94–2750.67 kPa for R-32 and 1745.26–2708.93 kPa for R-410A. The figure exhibits the mean HTC and FPD for CPs ranging from 1866.36–2725.45 kPa. The mean MF of these results is 24.427 kg/m²•s and the mean HF is 1.384 kW/m². Results R-410A, at a CP of 1779.75 kPa, obtained under similar conditions, are also displayed. Under comparable CP conditions, the mean HTC of R-32 was higher and the FPD was lower than that of R-410A.

Table 6 lists the thermal properties of saturated gas and liquid at different pressures. The TP results of R-32 and R-410A, obtained using the same test PHE, suggest that the differences shown in Figs. 1 to 3 between the two refrigerants may be due to their physical properties. As Table 6 demonstrates, R-32 has higher latent heat, specific heat capacity, and thermal conductivity, along with lower viscosity at higher pressures than R-410A. In TP flow, the liquid state's high thermal conductivity reduces the thermal resistance of the liquid film on the wall, thereby enhancing R-32's heat transfer performance compared to R-410A due to its higher latent heat and liquid thermal conductivity. Moreover, as the difference between liquid phase viscosity and the density difference between the liquid and gas phase increases, the FPD rises, as described by Ghiaasiaan [60], and is larger in R-410A as

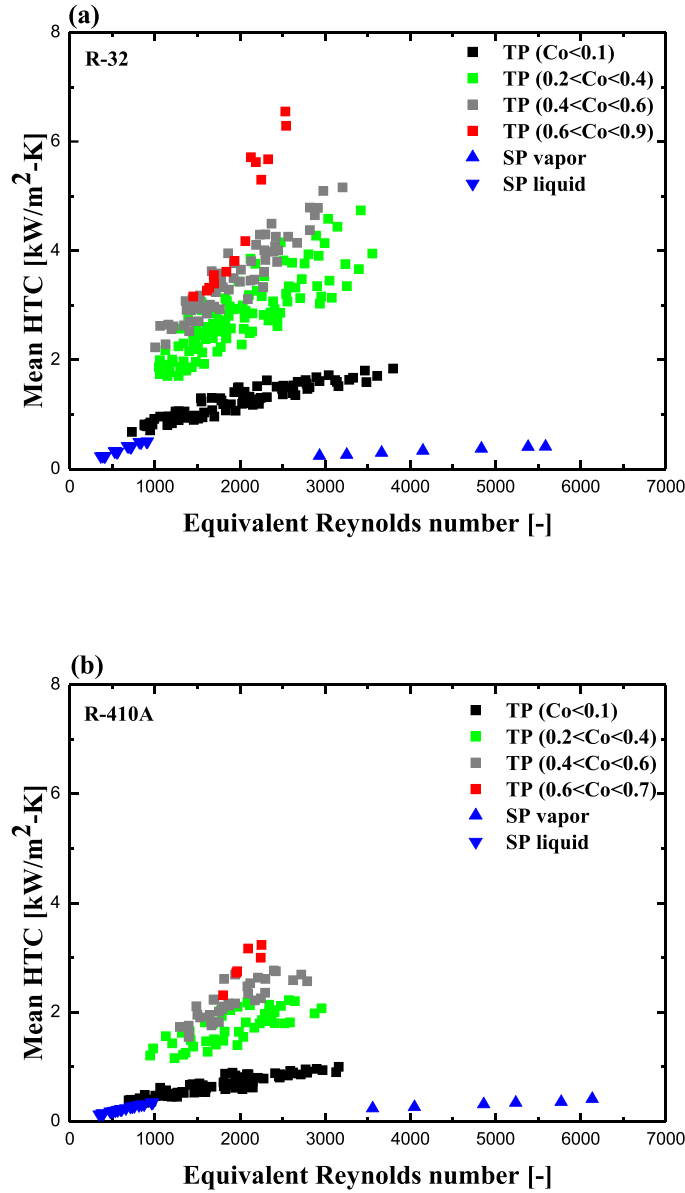


Fig. 11. Single- and two-phase mean HTCs of (a) R-32 and (b) R-410A with various Reynolds numbers.

indicated in Table 6.

3.2. Single-phase (SP) cooling results

Tables 7 and 8 provide summaries of the experimental conditions for the vapor and liquid cooling of SP R-32 and R-410A, respectively. The energy balance errors for the vapor and liquid, obtained under the conditions listed in Tables 7 and 8, are displayed in Fig. 9. The R-32 vapor cooling experiments recorded a total of seven data points within an energy balance error of $\pm 4.7\%$ across an average heat transfer rate range of 0.075–0.135 kW. The R-410A vapor cooling experiments recorded a total of six data points within an energy balance error of $\pm 3.6\%$ across an average heat transfer rate range of 0.062–0.078 kW. The R-32 liquid cooling experiments recorded sixteen data points within an energy balance error of $\pm 6.1\%$ across an average heat transfer rate range of 0.212–0.516 kW. The R-410A liquid cooling experiments recorded sixteen data points within an energy balance error of $\pm 4.2\%$ across an average heat transfer rate range of 0.110–0.359 kW.

Fig. 10 presents the SP vapor and liquid cooling experiment results

for R-32 and R-410A, respectively. Fig. 10 reveals that R-32's mean HTC was higher than that of R-410A, while R-32's FPD was lower. The lower temperature made SP R-32 HTC and FPD higher. Given that these experiments used the same test PHE, the differences between the two refrigerants can be attributed to their varying physical properties. As Table 6 indicates, R-32's specific heat capacity and thermal conductivity exceed those of R-410A in each state, which leads to a higher SP HTC for R-32, as validated by the SP HTC results. Moreover, the viscosity that influences SP FPD is more pronounced in R-410A than in R-32 for both gas and liquid phases, which is reflected in the SP FPD results.

3.3. Comparisons between phases and refrigerants

Figs. 11 and 12 depict the variations in mean HTC and FPD with changes in the Reynolds number (or the equivalent Reynolds number for TP flow). As shown in Figs. 11 and 12, both the mean HTC and FPD for TP flow surpass those of SP flow. According to the definition of the equivalent Reynolds number (Eqs. (1)–(3)), an increase in the mean VQ of the TP flow leads to a corresponding increase in the equivalent Reynolds number. In Figs. 11 and 12, it becomes apparent that the decline in mean HTC and FPD is more pronounced when transitioning from full evaporation to a super-heated vapor state in regions of high mean VQ, as opposed to the shift from full condensation to a sub-cooled liquid state in regions with low mean VQ.

Analyzing Figs. 11 and 12, it's clear that the HTC and FPD of SP liquid flow are lower than those of TP flow within comparable Reynolds number ranges. However, as the vapor ratio in TP flow intensifies (increasing the equivalent Reynolds number), both HTC and FPD of the TP flow ascend. Based on the trends presented and the physical properties listed in Table 6, the vapor state itself demonstrates lower HTC and FPD. However, in TP flow, a prominent shear effect arises, where the vapor component induces a swirling motion of the liquid flow along the PHE wall. This is identified as the primary contributor to the heightened HTC and FPD observed in TP flow, as corroborated by several studies including [15,43,44]. When observed on the same x- and y-axis scales, at comparable Reynolds numbers, the HTC for R-32 was found to be higher than that for R-410A, while the FPD for R-32 was lower compared to R-410A.

Examining the TP HTC results in Fig. 11, it is found that HTC increases as the Co rises. Additionally, the gradient of HTC, as influenced by changes in the equivalent Reynolds number for TP flow, also increases with Co. This occurs because the overall heat transfer coefficient increases as Co increases, consistent with the observations in Fig. 6. On the other hand, Fig. 12 shows that the FPD is not affected by Co but is strongly influenced by MF. This result aligns with those in Figs. 6(b) and 7(b).

3.4. Energy-saving performance evaluations

In order to evaluate the economic feasibility of the energy-saving aspect of the results, the EPP is calculated as Eq. (30) [44].

$$EPP = \frac{\dot{Q}}{\Delta \dot{W}_f} \quad (30)$$

The denominator of Eq. (30) is frictional pumping power loss ($\Delta \dot{W}_f$), which is calculated by Eq. (31).

$$\Delta \dot{W}_f = \frac{\dot{m} \cdot \Delta P_f}{\rho_m} \quad (31)$$

The mean density (ρ_m) in Eq. (31), is calculated as Eq. (32) for the TP flow.

$$\rho_m = x_{test,r,m} \cdot \rho_{r,v} + (1 - x_{test,r,m}) \cdot \rho_{r,l} \quad (32)$$

Fig. 18 show the graphs comparing the SP and TP EPPs of R-32 and R-410A from the results of this study. As shown in Fig. 13, the EPP of the

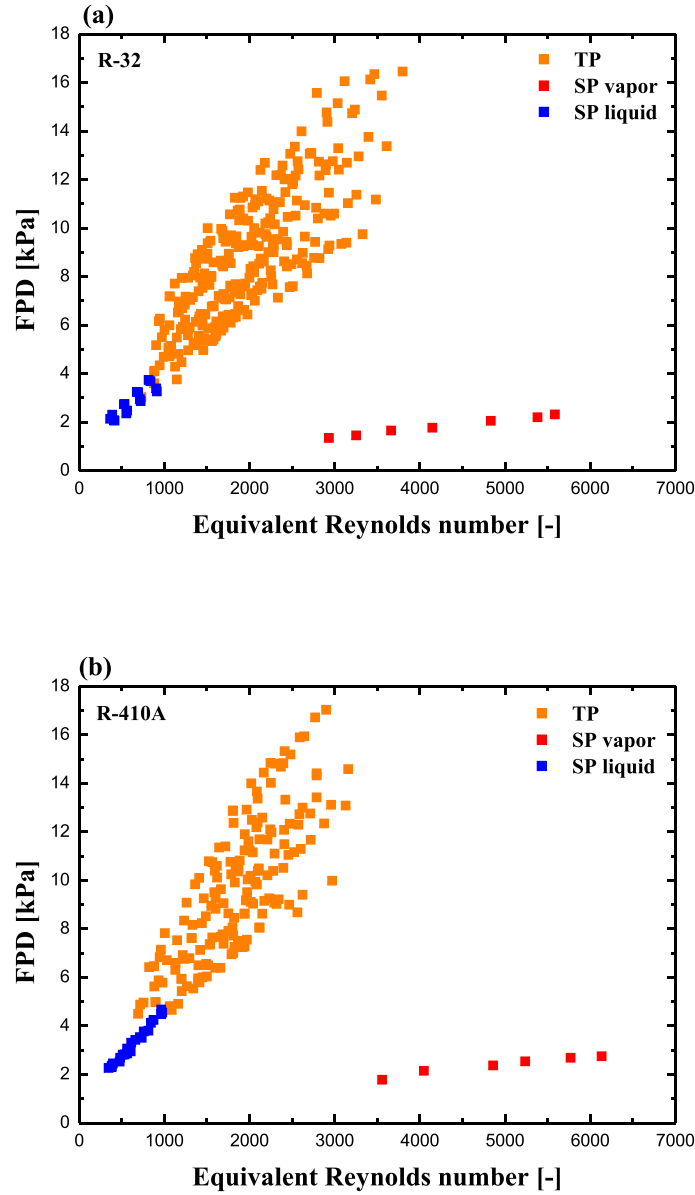


Fig. 12. Single- and two-phase FPDs of (a) R-32 and (b) R-410A with various Reynolds numbers.

TP flow is higher than that of the SP flow. Also, within the TP flow, as the Co increases, the EPP value increases, and when the Co is less than 0.1, the EPP value appears similar to that of the SP flow. When observed on the same x- and y-axis scales, at comparable Reynolds numbers, the EPP of R-32 was found to be higher than that for R-410A.

4. Correlations

4.1. TP condensation correlations

Based on the experimental data, correlations for the TP condensation Nu and f of R-32 and R-410A have been developed. These correlations consider the influences of HF, MF, CP, and VQ. The form of the TP Nu correlation developed in this study is presented in Eq. (33):

$$Nu_{tp} = C_1 \cdot Re_{eq}^{C_2} \cdot Co^{C_3} \cdot Pr_l^{0.3333} \quad (33)$$

The detailed specifications of the TP Nu correlations for each refrigerant are summarized in Table 9.

The form of the TP f correlation is presented in Eq. (34):

$$f_{tp} = C_1 \cdot Re_{eq}^{C_2} \cdot Re_l^{C_3} \quad (34)$$

The detailed specifications of the TP f correlations for each refrigerant are summarized in Table 10. Both Eqs. (33) and (34) can be applied within an error range of $\pm 10\%$, as demonstrated in Fig. 14.

Correlations for predicting the Nu and f of all TP data of integrated R-32 and R-410A were developed. The integrated R-32 and R-410A TP Nu correlation developed in this study is presented in Eq. (35):

$$Nu_{tp} = 0.2162 \cdot Re_{eq}^{0.7129} \cdot Co^{0.5426} \cdot Pr_l^{0.3333} \cdot \left(\frac{k_l}{k_v} \right)^{0.5599} \quad (35)$$

The detailed specifications of the integrated TP R-32 and R-410A Nu correlation are summarized in Table 11.

The integrated TP R-32 and R-410A f correlation is presented in Eq. (36):

$$f_{tp} = 1.312 \cdot 10^5 \cdot Re_{eq}^{-0.9224} \cdot Re_l^{-0.4181} \cdot \left(\frac{\rho_l}{\rho_v} \right)^{-0.2479} \quad (36)$$

The detailed specifications of the integrated TP R-32 and R-410A f

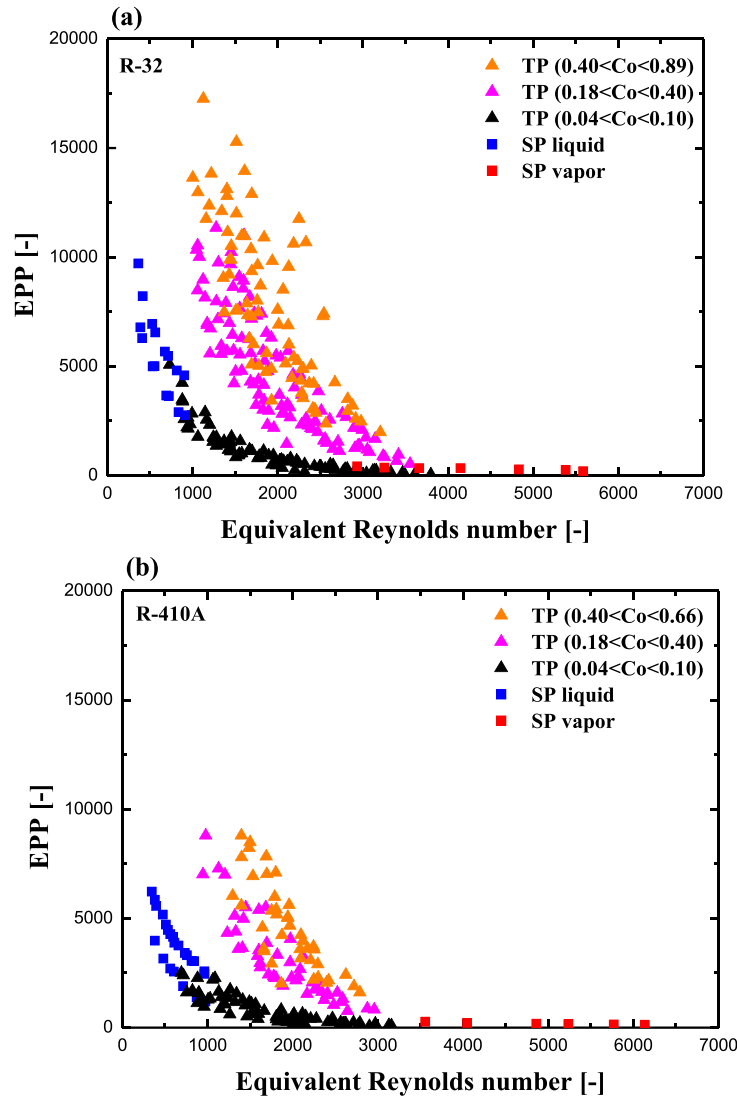


Fig. 13. Single- and two-phase EPP of (a) R-32 and (b) R-410A with various Reynolds numbers.

Table 9

Specifications for the TP Nu correlation of the refrigerants (Eq. (33)).

Refrigerants	C_1	C_2	C_3	Re_{eq} range	Co range	Pr_l range
R-32	0.4514	0.7442	0.5059	732.1~3797	0.042~0.889	1.855~1.988
R-410A	0.9273	0.6316	0.5927	694.2~3160	0.044~0.658	2.378~2.608

Table 10

Specifications for the TP f correlation of the refrigerants (Eq. (34)).

Refrigerants	C_1	C_2	C_3	Re_{eq} range	Re_l range
R-32	$3.155 \cdot 10^4$	-0.9001	-0.3386	732.1~3797	562.5~1082
R-410A	$3.899 \cdot 10^4$	-0.8955	-0.3403	694.2~3160	592.2~1107

correlation are summarized in Table 12. Through Eq. (35), 93.59 % of the total TP Nu data was predicted within ± 20 % error range, as shown in Fig. 15(a). Through Eq. (36), 95.88 % of the total TP f data was predicted within ± 20 % error range, as shown in Fig. 15(b).

4.2. SP cooling correlations

Based on the experimental data, correlations for the SP vapor and

liquid Nu and f of R-32 and R-410A have been developed, reflecting the influence of Re . The form of the SP Nu correlation developed in this study is presented in Eq. (37):

$$Nu_{sp} = C_1 \cdot Re_{sp}^{C_2} \cdot Pr_{sp}^{0.3333} \quad (37)$$

The detailed specifications of the SP Nu correlations for each refrigerant are summarized in Table 13.

The form of the SP f correlation is presented in Eq. (38):

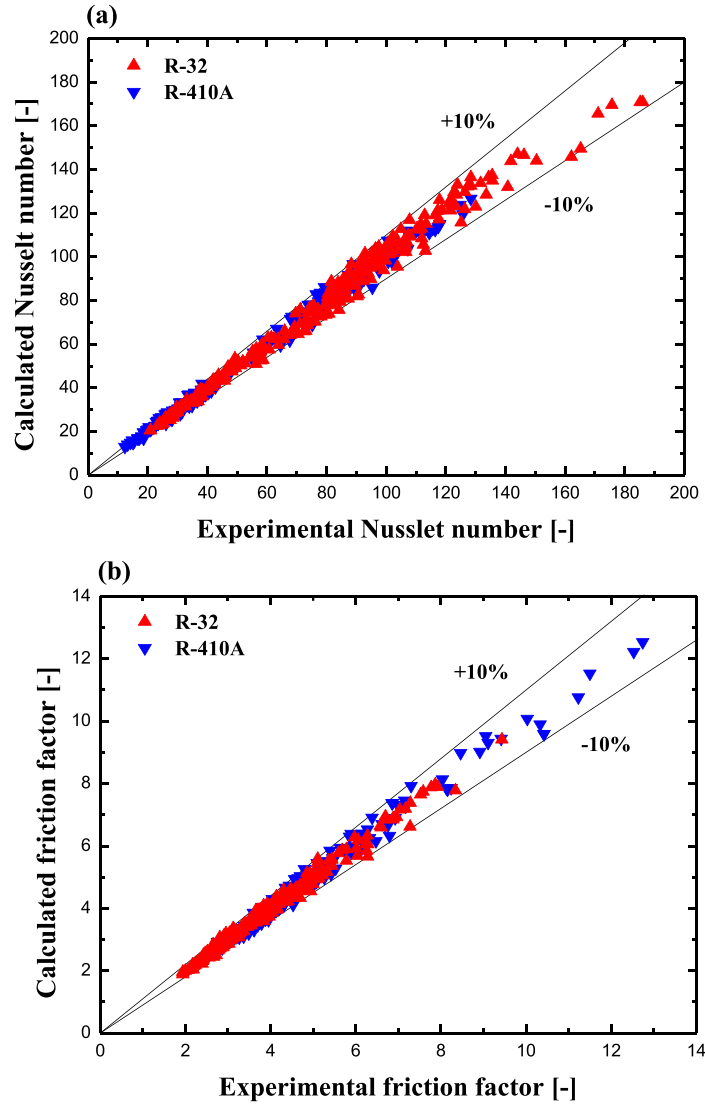


Fig. 14. Comparisons between experimental and calculated TP condensation (a) Nusselt numbers and (b) friction factors.

Table 11
Specifications for the TP Nu correlation of the refrigerants (Eq. (35)).

Refrigerant	Re_{eq} range	Co range	Pr_1 range	$\left(\frac{k_l}{k_v}\right)$ range
R-32 & R-410A	694.2~3797	0.042~0.889	1.855~2.609	3.930~6.855

Table 12
Specifications for the TP f correlation of the refrigerants (Eq. (36)).

Refrigerant	Re_{eq} range	Re_l range	$\left(\frac{\rho_l}{\rho_v}\right)$ range
R-32 & R-410A	694.2~3797	562.5~1107	7.938~19.13

$$f_{sp} = C_1 \cdot Re_{sp}^{C_2} \quad (38)$$

The detailed specifications of the SP f correlations for each refrigerant are summarized in Table 14. The applicable error ranges for Eqs. (37) and (38) are depicted in Fig. 16.

The correlations for predicting the Nu and f of all SP data of integrated R-32 and R-410A were developed. The integrated SP R-32 and R-410A Nu correlation developed in this study is presented in Eq. (39):

$$Nu_{sp} = 0.01855 \cdot Re_{sp}^{0.9306} \cdot Pr_{sp}^{0.3333} \quad (39)$$

The detailed specifications of the SP Nu correlations for each refrigerant are summarized in Table 15.

The integrated SP R-32 and R-410A f correlation is presented in Eq. (40):

$$f_{sp} = 7.665 \cdot 10^4 \cdot Re_{sp}^{-1.347} \cdot \left(\frac{\mu}{\mu_l}\right)^{-0.08339} \quad (40)$$

The detailed specifications of the SP f correlations for each refrigerant are summarized in Table 16. Eq. (39) was applicable within the $\pm 20\%$ error range and is shown in Fig. 17(a). Eq. (40) predicted 96.43 % of the total SP data within the $\pm 20\%$ error range and is shown in Fig. 17 (b).

4.3. Comparisons with other studies

Fig. 18 presents a comparison of the R-32 correlations from this study with those developed in the previous research. The experimental results from this study were applied to each of the correlations, facilitating a comparison between the experimental and the calculated values. Details regarding the experimental conditions and correlations from other

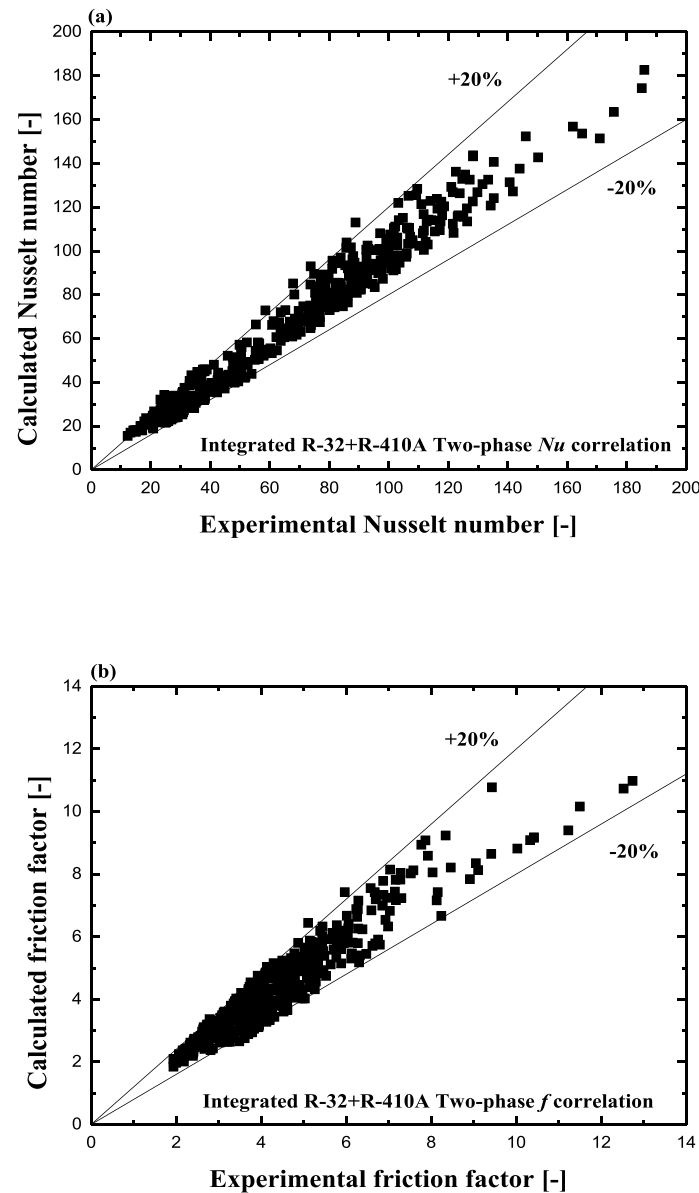


Fig. 15. Comparisons between experimental and calculated integrated TP R-32 & R-410A condensation (a) Nusselt numbers and (b) friction factors.

Table 13

Specifications for the SP Nu correlation of the refrigerants.

Refrigerants	States	C_1	C_2	Re range	Pr range
R-32	Vapor	0.07109	0.7775	2935~6311	1.414~1.446
	Liquid	0.01801	0.9495	364.9~911.9	1.841~1.861
R-410A	Vapor	0.01068	0.9907	3189~6136	1.678~1.696
	Liquid	0.0151	0.9477	346.1~975.4	2.374~2.384

Table 14

Specifications for the SP f correlation of the refrigerants.

Refrigerants	States	C_1	C_2	Re range
R-32	Vapor	$1.982 \cdot 10^4$	-1.182	2935~6311
	Liquid	$6.891 \cdot 10^4$	-1.352	364.9~911.9
R-410A	Vapor	$4.320 \cdot 10^4$	-1.238	3189~6136
	Liquid	$8.334 \cdot 10^4$	-1.346	346.1~975.4

studies, depicted in Fig. 18, are summarized in Table 17. Table 17 systematically arranges information regarding the experimental conditions, PHE configurations, correlations, and definitions of dimensionless numbers associated with the correlations, for both this study and previous research.

In Fig. 18(a), a comparison is made between the experimental Nu from the present study and the Nu values calculated using correlations from Table 17. Among these, Lee et al.'s Nu correlation [61] accurately predicted 27.61 % of the Nu data within a ± 10 % error range. However, despite Lee et al. [61] conducting experiments at lower temperatures, which typically suggest higher Nu predictions, the smaller β and b in their test PHE [61] led to lower Nu predictions for this study [62,63]. Similarly, Rossato et al. [64], while having experimental conditions closely matching this study except for CP differences, employed the Nu correlation by Amalfi et al. [65]. Their experiments, conducted at a lower CP, would normally imply higher Nu predictions, yet the smaller β and b in Rossato et al.'s PHE [64] resulted in a lower Nu prediction than Lee et al. [61].

Further, as indicated in Fig. 18(b), Lee et al.'s f correlation [61] did not accurately predict any f data points within a ± 10 % error range. The

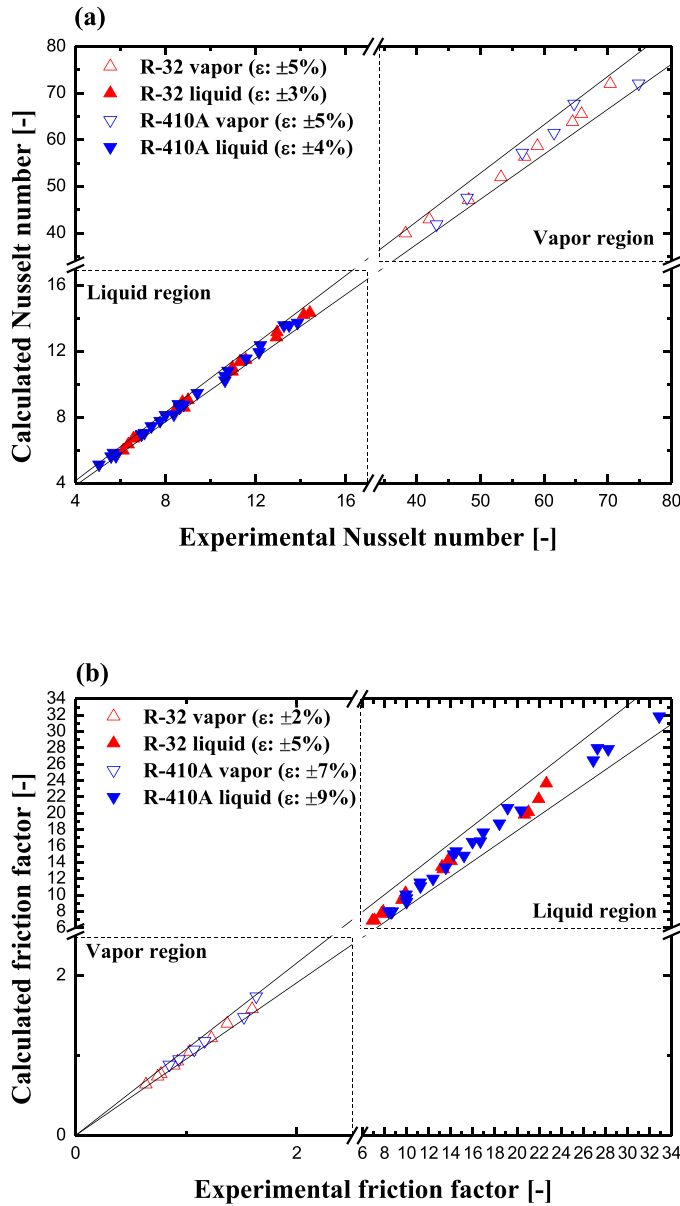


Fig. 16. Comparisons between experimental and calculated SP (a) Nusselt numbers and (b) friction factors of R-32 and R-410A.

Table 15
Specifications for the SP Nu correlation of the refrigerants (Eq. (39)).

Refrigerants	States	Re range	Pr range
R-32 & R-410A	Vapor & Liquid	346.1~6311	1.414~2.384

Table 16
Specifications for the SP f correlation of the refrigerants (Eq. (40)).

Refrigerants	States	Re range
R-32 & R-410A	Vapor & Liquid	346.1~6311

discrepancies evident in Fig. 18(b) are attributable to variations in the test PHEs used in each study. Notably, the f value predicted by Lee et al.'s correlation [61] was lower than the experimental value observed in this study. Lee et al. [61] developed their f correlation using a PHE shorter than the one used in this study, leading to an underestimation of

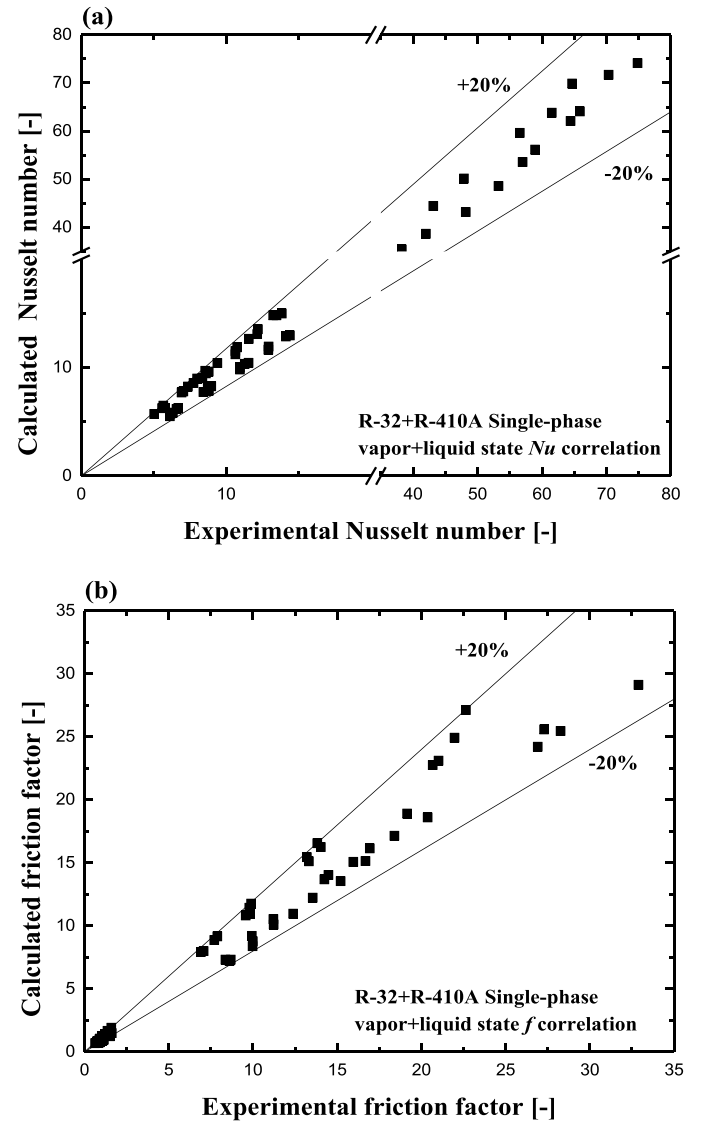


Fig. 17. Comparisons between experimental and calculated integrated SP R-32 and R-410A (a) Nusselt numbers and (b) friction factors.

the f value.

As shown in Fig. 19(a), applying dimensionless numbers from this study's results, Hsieh and Lin's evaporation correlation [25] predicted a higher Nu compared to Han et al. [35] (evaporation) and Kuo et al. [36] (condensation), although it overestimated this study's Nu . A reduced λ or an increased b , as indicated in Refs. [62,63], elevates the Nu by intensifying the refrigerant's secondary vortex flow. Hsieh and Lin's λ [25] is about twice that of Han et al. [35], leading to a higher Nu prediction. Han et al. [35] conducted experiments under similar MF and HF but reduced CP conditions compared to this study. Despite these parameters suggesting a higher Nu , the Nu calculated from Han et al.'s correlation [35] was lower than the experimental value here, emphasizing that b affects Nu more than λ . Variations between this study and Han et al. [35] in λ and b were less significant than with Hsieh and Lin [25]. Ham et al. [66] noted that increasing L_{width} raises HTC, and since L_{width} in this study is greater than in Han et al. [35], it resulted in a higher Nu .

Kuo et al. [36], operating under higher MF and HF but lower CP conditions and with a larger corrugation furrow area, calculated a Nu lower than our experimental value, likely due to their lower β causing weaker turbulence. Both Hsieh and Lin [25] and Kuo et al. [36] used similar PHEs as this study but at significantly lower CP, with Hsieh and

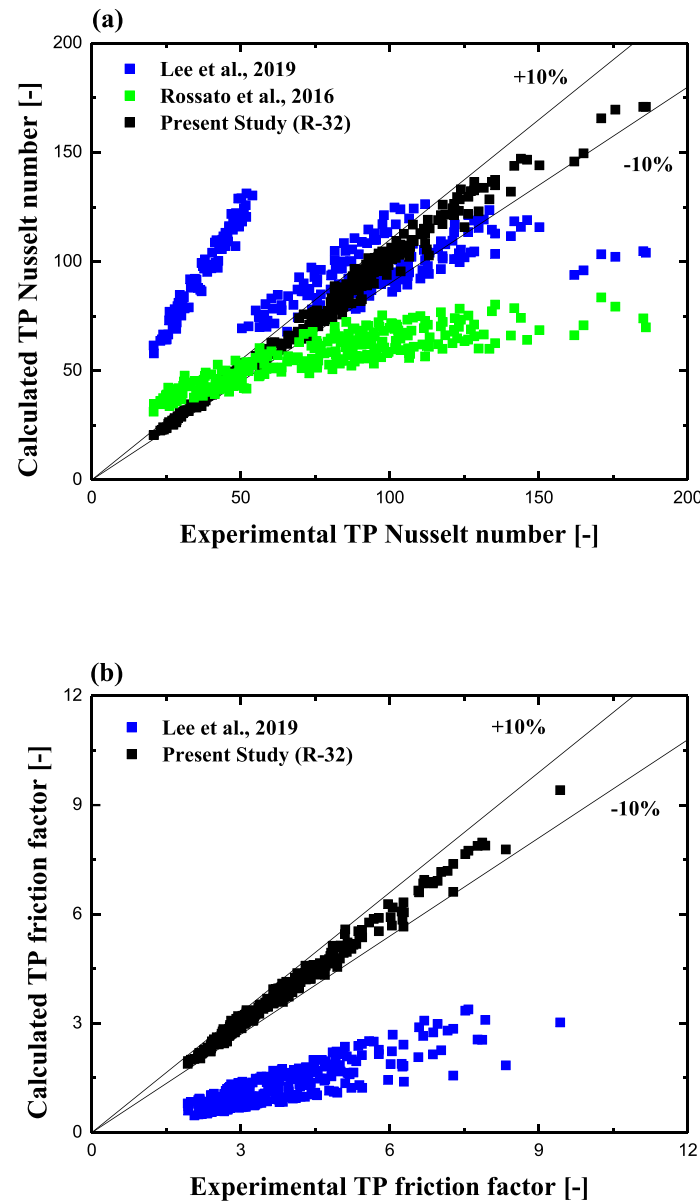


Fig. 18. Comparisons between experimental and calculated SP vapor-liquid cooling (a) Nusselt numbers and (b) friction factors.

Lin [25] operating at the lowest CP, resulting in their correlation predicting a higher Nu . Fig. 19(b) shows that correlations of Hsieh and Lin [25] and Kuo et al. [36] predicted over 50 % of f values within $\pm 10\%$ error. Their studies [25,36] demonstrate that TP HTC increases with decreasing pressure (or temperature) in the same PHE, with Hsieh and Lin's [25] lower pressure evaluation leading to higher HTC and FPD predictions than Kuo et al. [36]. Changes in MF primarily affect FPD, while HF and CP changes have minor impacts, leading Hsieh and Lin's [25] correlation to predict slightly higher f than Kuo et al. [36].

All three studies [25,35,36] used vertically elongated PHEs, more than twice as long as this study's PHE. Thus, while Nu correlations by Han et al. [35] and Kuo et al. [36] predicted lower Nu than our experimental results, their f correlations consistently resulted in higher f values than observed in this experiments [67].

5. Conclusions

In this study, the TP condensation, SP vapor and liquid cooling heat transfer, and frictional pressure drop characteristics of R-32 and R-410A were investigated. Based on the experimental results, the EPP for all R-

32 and R-410A phases was assessed, and correlations for the Nu and f were developed. The experimental results and EPP of R-32 were compared with those of R-410A, and the developed correlations were contrasted with those from other research. The key conclusions drawn from this study are as follow:

- (1) In this study, the changes in TP condensation HTCs and FPDs of R-32 and R-410A based on the variations in HF, MF, CP, and VQ were evaluated. Similarly, the changes in SP HTCs and FPDs of R-32 and R-410A were assessed based on temperature and flow rate variations. The TP HTC and FPD increased with the increase in MF and VQ and decreased with the reduction in CP. An increase in HF led to an increase in TP HTC but had a negligible effect on TP FPD. The SP HTC and FPD increased with increasing flow rate and lowering of temperature. Overall, under similar conditions, the thermal and hydraulic performance of R-32 TP condensation was found to be superior to that of R-410A.
- (2) Based on the experimental results of TP and SP flows of R-32 and R-410A in this study, the EPP s were evaluated. The EPP was higher in TP than in SP, and it was higher when the Reynolds

Table 17
R-32 TP correlations from other studies.

Authors	Experimental conditions PHE parameters	Correlations
Lee et al. [61]	G: 45 kg/m²•s q'': 6.45 kW/m² T: 15 °C P: 1281 kPa R-32 Evap. L_{port}: 0.234 m D_{port}: 0.033 m L_{width}: 0.117 m λ: 0.0075 m t: 0.0004 m b: 0.00194 m β: 60°	$Nu_{tp} = 1.84 \cdot Re_{eq}^{0.58} \cdot Re_{lo}^{0.22} \cdot Pr_l^{1.59}$ $Nu_{tp} = 1.84 \cdot Re_{eq}^{0.58} \cdot Re_{lo}^{0.22} \cdot Pr_l^{1.59}$ $f_{tp} = 4.77 \cdot 10^6 \cdot Re_{eq}^{-0.96} \cdot Re_{lo}^{-0.24} \cdot Pr_l^{-9.77}$
Rossato et al. [64]	G: 18–80 kg/m²•s q'': 3–27 kW/m² T: 5 °C P: 951.5 kPa R-32 Evap. L_{tot}: 0.529 m L_{width}: 0.113 m λ: 0.004573 m b: 0.00146 m β: 60°	$Nu_{tp} = 18.495 \cdot (\beta^*)^{0.248} \cdot Re_v^{0.135} \cdot Bd^{0.235} \cdot Bo^{0.198} \cdot (\rho^*)^{-0.223}$ from Amalfi et al. [65] $\beta^* = \frac{\beta}{70}$ $Bd = \frac{(\rho_l - \rho_v) \cdot g \cdot D_{hyd}^2}{\sigma_s}$ (σ_s: Surface tension, N/m) $Bo = \frac{q''}{G \cdot i_{fg}}$ $\rho^* = \frac{\rho_l}{\rho_v}$
Hsieh and Lin [25]	G: 50–200 kg/m²•s $T_{r,i}$: 8–15 °C R-410A Liq. Heating L_{port}: 0.45 m D_{port}: 0.025 m L_{width}: 0.12 m λ: 0.01 m t: 0.0004 m b: 0.0029 m β: 60°	$h_l = 0.2092 \cdot \frac{k_l}{D_{hyd}} \cdot Re_l^{0.78} \cdot Pr_l^{1/3} \cdot \left(\frac{\mu_m}{\mu_{wall}} \right)^{0.14}$
Hsieh and Lin [25]	G: 50–125 kg/m²•s q'': 5–35 kW/m² P_{sat}: 1.08–1.44 MPa R-410A Evap. L_{port}: 0.45 m D_{port}: 0.025 m L_{width}: 0.12 m λ: 0.01 m t: 0.0004 m b: 0.0029 m β: 60°	$h_{tp} = h_l \cdot (88 \cdot Bo^{0.5})$ $f_{tp} = 61000 \cdot Re_{eq}^{-1.25}$ (Upward flow) $Bo = \frac{q''}{G \cdot i_{fg}}$
Han et al. [35]	G: 13–34 kg/m²•s q'': 2.5–8.5 kW/m² T_{evap}: 5–15 °C R-410A Evap. L_{port}: 0.476 m D_{port}: 0.0195 m L_{width}: 0.115 m λ: 0.0052 m t: 0.0004 m b: 0.00215 m β: 65°	$Nu_{tp} = Ge_1 \cdot Re_{eq}^{Ge_2} \cdot Bo_{eq}^{0.3} \cdot Pr^{0.4}$ $f_{tp} = Ge_3 \cdot Re_{eq}^{Ge_4}$ (Downward flow) $Ge_1 = 2.81 \cdot \left(\frac{\lambda}{D_{hyd}} \right)^{-0.041} \cdot \left(\frac{\pi}{2} - \beta \right)^{-2.83}$ $Ge_2 = 0.746 \cdot \left(\frac{\lambda}{D_{hyd}} \right)^{-0.082} \cdot \left(\frac{\pi}{2} - \beta \right)^{0.61}$ $Bo_{eq} = \frac{q''}{G_{eq} \cdot i_{fg}}$ $Ge_3 = 64,710 \cdot \left(\frac{\lambda}{D_{hyd}} \right)^{-5.27} \cdot \left(\frac{\pi}{2} - \beta \right)^{-3.03}$ $Ge_4 = -1.314 \cdot \left(\frac{\lambda}{D_{hyd}} \right)^{-0.62} \cdot \left(\frac{\pi}{2} - \beta \right)^{-0.47}$
Kuo et al. [36]	G: 50–150 kg/m²•s q'': 5–20 kW/m² P_{sat}: 1.44–1.95 MPa R-410A Cond. L_{port}: 0.45 m D_{port}: 0.025 m L_{width}: 0.12 m λ: 0.01 m t: 0.0004 m b: 0.0029 m β: 60°	$h_{tp} = h_l \cdot [0.25 \cdot Co^{-0.45} \cdot Fr_l^{0.25} + 75 \cdot Bo^{0.75}]$ $f_{tp} = 21500 \cdot Re_{eq}^{-1.14} \cdot Bo^{-0.085}$ (Downward flow) $Co = \left(\frac{v_l}{v_g} \right) \cdot \left(\frac{1 - x_m}{x_m} \right)^{0.8}$ $Fr_l = \frac{G}{\rho_l^2 \cdot g \cdot D_{hyd}}$

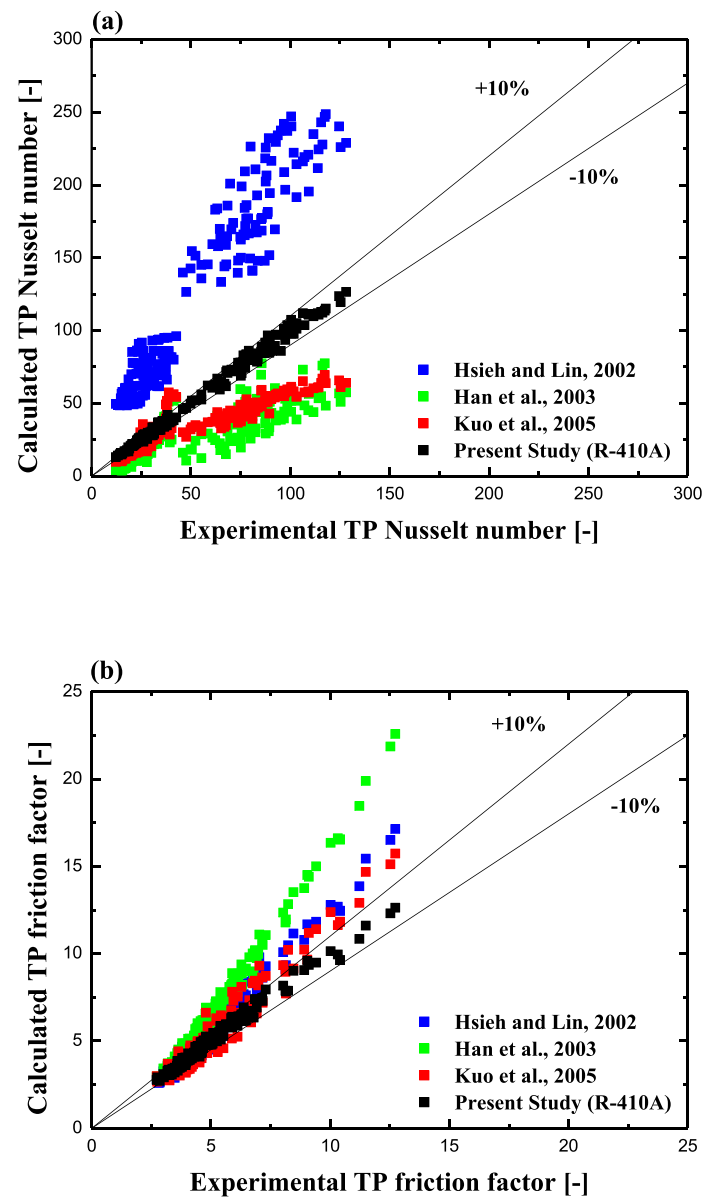


Fig. 19. Comparisons between experimental and calculated SP vapor-liquid cooling (a) Nusselt numbers and (b) friction factors.

number was lower. The SP *EPP* was higher in the liquid state than that in the vapor state, while the TP *EPP* increased as the *Co* increased. Overall, in all phases and across all Reynolds number ranges, the *EPP* of R-32 was higher than that of R-410A.

- (3) Based on the experimental results, R-32 and R-410A correlations for the TP condensation and SP liquid and vapor *Nu* and *f* were developed. The TP condensation correlations were applicable within $\pm 10\%$ error range. The error range for the SP *Nu* correlation was most pronounced for vapor R-32 and vapor R-410A at $\pm 5\%$, while the error range for the SP *f* correlation was largest for vapor R-410A at $\pm 9\%$. The integrated TP and SP R-32 and R-410A *Nu* and *f* correlations were developed and predicted more than 90 % of total data within $\pm 20\%$ error range.
- (4) In comparing the R-32 and R-410A TP condensation correlations developed in this study with those from other studies, it was found that the correlations from other research were unsuitable for predicting the HTC and FPD of this study. From the comparisons with other studies, it was confirmed that not only the refrigerant temperature but also overall PHE geometry factors,

such as the chevron angle and corrugation depth, influence the *Nu* and *f*.

CRediT authorship contribution statement

Joo Young Song: Writing – original draft, Methodology, Investigation, Conceptualization. **Dong Ho Kim:** Methodology, Investigation, Formal analysis. **Chan Ho Song:** Methodology, Investigation, Data curation. **Yong Tae Kang:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

This work was supported by the Korea Institute of Energy Technology Evaluation and Planning (KETEP) and the Ministry of Trade, Industry & Energy (MOTIE) of the Republic of Korea (No. 00239724).

References

- [1] M. Protocol, Montreal Protocol On Substances That Deplete the Ozone Layer, 26, US Government Printing Office, Washington, DC, 1987, pp. 128–136.
- [2] M. Schulz, D. Kourkoulas, Regulation (EU) No 517/2014 of The European Parliament and of the council of 16 April 2014 on fluorinated greenhouse gases and repealing Regulation (EC) No 842/2006, Off. J. Eur. Union 2014 (517) (2014) L150.
- [3] E.A. Heath, Amendment to the Montreal protocol on substances that deplete the ozone layer (Kigali amendment), Int. Legal Mater. 56 (1) (2017) 193–205.
- [4] P. Agreement, Paris agreement, in: Report of the Conference of the Parties to the United Nations Framework Convention on Climate Change (21st Session, 2015: Paris), 2015, p. 2017. Retrived December, HeinOnline.
- [5] J. Hughes, J. Juhasz, Low GWP Refrigerant Solutions For Heat Pumps, Chemours Co, 2019.
- [6] Í.F. Guilherme, D.F.M. Pico, D.D.O. dos Santos, E.P. Bandarra Filho, A review on the performance and environmental assessment of R-410A alternative refrigerants, J. Build. Eng. 47 (2022) 103847.
- [7] M. Petersen, K. Steve, Review of lower GWP refrigerants for retrofitting R-410A applications, in: 19th International Refrigeration and Air Conditioning Conference, 2022.
- [8] B. Kim, S.H. Lee, D. Lee, Y. Kim, Performance comparison of heat pumps using low global warming potential refrigerants with optimized heat exchanger designs, Appl. Therm. Eng. 171 (2020) 114990.
- [9] V.H. Panato, D.F.M. Pico, E.P. Bandarra Filho, Experimental evaluation of R32, R452B and R454B as alternative refrigerants for R410A in a refrigeration system, Int. J. Refrig. 135 (2022) 221–230.
- [10] F. Pardo, G. Zarca, A. Urtiaga, Effect of feed pressure and long-term separation performance of Pebax-ionic liquid membranes for the recovery of difluoromethane (R32) from refrigerant mixture R410A, J. Memb. Sci. 618 (2021) 118744.
- [11] R.P. Ribeiro, J.E. Sosa, J.M. Araújo, A.B. Pereira, J.P. Mota, Vacuum swing adsorption for R-32 recovery from R-410A refrigerant blend, Int. J. Refrig. (2023).
- [12] G.A. Longo, A. Gasparella, Refrigerant R134a vaporisation heat transfer and pressure drop inside a small brazed plate heat exchanger, Int. J. Refrig. 30 (5) (2007) 821–830.
- [13] Y. Hsieh, T. Lin, Evaporation heat transfer and pressure drop of refrigerant R-410A flow in a vertical plate heat exchanger, J. Heat Transf. 125 (5) (2003) 852–857.
- [14] Y. Yan, T. Lin, Evaporation heat transfer and pressure drop of refrigerant R-134a in a plate heat exchanger, J. Heat Transf. 121 (1) (1999) 118–127.
- [15] O.J. Kwon, B. Shon, Y.T. Kang, Experimental investigation on condensation heat transfer and pressure drop of a low GWP refrigerant R-1233zd (E) in a plate heat exchanger, Int. J. Heat Mass Transf. 131 (2019) 1009–1021.
- [16] T.S. Khan, M. Khan, M.-C. Chyu, Z. Ayub, Experimental investigation of single phase convective heat transfer coefficient in a corrugated plate heat exchanger for multiple plate configurations, Appl. Therm. Eng. 30 (8–9) (2010) 1058–1065.
- [17] J. Yang, A. Jacobi, W. Liu, Heat transfer correlations for single-phase flow in plate heat exchangers based on experimental data, Appl. Therm. Eng. 113 (2017) 1547–1557.
- [18] B.P. Rao, B. Sundén, S.K. Das, An experimental and theoretical investigation of the effect of flow maldistribution on the thermal performance of plate heat exchangers, J. Heat Transf. 127 (3) (2005) 332–343.
- [19] A. Muley, R. Manglik, Experimental study of turbulent flow heat transfer and pressure drop in a plate heat exchanger with chevron plates, J. Heat Transf. 121 (1) (1999) 110–117.
- [20] K. Nilpueng, S. Wongwises, Experimental study of single-phase heat transfer and pressure drop inside a plate heat exchanger with a rough surface, Exp. Therm. Fluid Sci. 68 (2015) 268–275.
- [21] A. Bhattad, J. Sarkar, P. Ghosh, Discrete phase numerical model and experimental study of hybrid nanofluid heat transfer and pressure drop in plate heat exchanger, Int. Commun. Heat Mass Transf. 91 (2018) 262–273.
- [22] T.-P. Teng, T.-C. Hsiao, C.-C. Chung, Characteristics of carbon-based nanofluids and their application in a brazed plate heat exchanger under laminar flow, Appl. Therm. Eng. 146 (2019) 160–168.
- [23] J.Y. Song, J.H. Park, Y.T. Kang, Heat transfer and frictional pressure drop characteristics of H₂O/LiBr solution in plate heat exchangers for triple-effect absorption application, Appl. Therm. Eng. 189 (2021) 116730.
- [24] J.Y. Song, J.W. Lee, Y.T. Kang, Comparisons of Nu correlations for H₂O/LiBr solution in plate heat exchanger for triple effect absorption chiller application, Energy 172 (2019) 852–860.
- [25] Y. Hsieh, T. Lin, Saturated flow boiling heat transfer and pressure drop of refrigerant R-410A in a vertical plate heat exchanger, Int. J. Heat Mass Transf. 45 (5) (2002) 1033–1044.
- [26] M. Khoshvaght-Aliabadi, P. Ghodrati, H. Mortazavi, Y.T. Kang, Predicting heat transfer and flow features of supercritical CO₂ in printed circuit heat exchangers with novel wavy minichannels, Int. J. Heat Mass Transf. 196 (2022) 123232.
- [27] M. Khoshvaght-Aliabadi, P. Ghodrati, O. Mahian, Y.T. Kang, Enhancing the performance of supercritical CO₂-cooled heat sink by converging-diverging minichannels, Appl. Therm. Eng. 234 (2023) 121225.
- [28] M. Khoshvaght-Aliabadi, P. Ghodrati, Y.T. Kang, Optimal combination of converging and diverging minichannels in PCHE as pre-cooler under diverse operating conditions of supercritical CO₂, Energy 272 (2023) 127158.
- [29] M. Khoshvaght-Aliabadi, P. Ghodrati, Y.T. Kang, Thermohydraulic characteristics of printed circuit recuperators in a supercritical CO₂ brayton cycle with nonuniform minichannels, Int. J. Energy Res. (2023) 2023.
- [30] M. Khoshvaght-Aliabadi, P. Ghodrati, H. Mortazavi, Y.T. Kang, Numerical analysis of heat transfer and flow characteristics of supercritical CO₂-cooled wavy minichannel heat sinks, Appl. Therm. Eng. 226 (2023) 120307.
- [31] C. Park, S. An, T. Kim, Y.-I. Kim, A. Aldabahi, M. El-Newehy, S.S. Yoon, Pool boiling enhancement via biphilic surface comprising superhydrophilic TiO₂ and superhydrophobic Teflon arrays, Int. J. Heat Mass Transf. 202 (2023) 123675.
- [32] C. Park, B.Y. Kim, Y.-I. Kim, Y.T. Kang, S.S. Yoon, Pool boiling performance of TiO₂ superhydrophilic and Teflon superhydrophobic surfaces on evenly deposited copper frustums, Case Stud. Therm. Eng. 41 (2023) 102581.
- [33] Y.I. Kim, B.-H. Bang, K. Jang, S. An, A.L. Yarin, S.S. Yoon, Pool boiling enhancement via nanotexturing and self-propelled swing motion for bubble shedding, Int. Commun. Heat Mass Transf. 133 (2022) 105934.
- [34] Y.M. Ko, J.Y. Song, J.W. Lee, S. Sohn, C.H. Song, M. Khoshvaght-Aliabadi, Y. Kim, Y.T. Kang, A critical review on Colburn j-factor and f-factor and energy performance analysis for finned tube heat exchangers, Energy 287 (2024) 129609.
- [35] D.-H. Han, K.-J. Lee, Y.-H. Kim, Experiments on the characteristics of evaporation of R410A in brazed plate heat exchangers with different geometric configurations, Appl. Therm. Eng. 23 (10) (2003) 1209–1225.
- [36] W. Kuo, Y. Lie, Y. Hsieh, T. Lin, Condensation heat transfer and pressure drop of refrigerant R-410A flow in a vertical plate heat exchanger, Int. J. Heat Mass Transf. 48 (25–26) (2005) 5205–5220.
- [37] G.A. Longo, Heat transfer and pressure drop during HFC refrigerant saturated vapour condensation inside a brazed plate heat exchanger, Int. J. Heat Mass Transf. 53 (5–6) (2010) 1079–1087.
- [38] G.A. Longo, R410A condensation inside a commercial brazed plate heat exchanger, Exp. Therm. Fluid Sci. 33 (2) (2009) 284–291.
- [39] G.A. Longo, S. Mancin, G. Righetti, C. Zilio, Local heat transfer coefficients of R32 and R410a condensation inside a brazed plate heat exchanger (BPHE), Int. J. Heat Mass Transf. 194 (2022) 123041.
- [40] G.A. Longo, S. Mancin, G. Righetti, C. Zilio, Local heat transfer coefficients of R32 and R410A boiling inside a brazed plate heat exchanger (BPHE), Appl. Therm. Eng. 215 (2022) 118930.
- [41] G.A. Longo, Refrigerant R134a condensation heat transfer and pressure drop inside a small brazed plate heat exchanger, Int. J. Refrig. 31 (5) (2008) 780–789.
- [42] G.A. Longo, C. Zilio, G. Righetti, J.S. Brown, Experimental assessment of the low GWP refrigerant HFO-1234ze (Z) for high temperature heat pumps, Exp. Therm. Fluid Sci. 57 (2014) 293–300.
- [43] P. Vlasogiannis, G. Karagiannis, P. Argyropoulos, V. Bontozoglou, Air–water two-phase flow and heat transfer in a plate heat exchanger, Int. J. Multiphase Flow 28 (5) (2002) 757–772.
- [44] J.H. Jung, Y.M. Ko, Y.T. Kang, Condensation heat transfer characteristics and energy conversion performance analysis for low GWP refrigerants in plate heat exchangers, Int. J. Heat Mass Transf. 166 (2021) 120727.
- [45] O.J. Kwon, J.H. Jung, Y.T. Kang, Development of experimental Nusselt number and friction factor correlations for condensation of R-1233zd (E) in plate heat exchangers, Int. J. Heat Mass Transf. 158 (2020) 120008.
- [46] Y.M. Ko, J.H. Jung, S. Sohn, C.H. Song, Y.T. Kang, Condensation heat transfer performance and integrated correlations of low GWP refrigerants in plate heat exchangers, Int. J. Heat Mass Transf. 177 (2021) 121519.
- [47] H. Martin, A theoretical approach to predict the performance of chevron-type plate heat exchangers, Chem. Eng. Process. Process Intensif. 35 (4) (1996) 301–310.
- [48] E. Wilson, A basis for rational design of heat transfer apparatus, J. Heat Transf. 37 (1915) 546–551.
- [49] J. Fernández-Seara, F.J. Uhía, J. Sieres, A. Campo, A general review of the Wilson plot method and its modifications to determine convection coefficients in heat exchange devices, Appl. Therm. Eng. 27 (17–18) (2007) 2745–2757.
- [50] W. Nusselt, Die oberflächenkondensation des wasserdampfes, VDI-Zs 60 (1916) 541.
- [51] D. Lee, S. Yun, J. Choi, Y. Kim, Flow patterns and heat transfer characteristics of R-1234ze (E) for downward condensation in a plate heat exchanger, Int. J. Heat Mass Transf. 175 (2021) 121373.
- [52] X. Tao, C.A.I. Ferreira, NH₃ condensation in a plate heat exchanger: flow pattern based models of heat transfer and frictional pressure drop, Int. J. Heat Mass Transf. 154 (2020) 119774.
- [53] X. Tao, E. Dahlgren, M. Leichsenring, C.A.I. Ferreira, NH₃ condensation in a plate heat exchanger: experimental investigation on flow patterns, heat transfer and frictional pressure drop, Int. J. Heat Mass Transf. 151 (2020) 119374.
- [54] A. Jokar, M.H. Hosni, S.J. Eckels, Dimensional analysis on the evaporation and condensation of refrigerant R-134a in minichannel plate heat exchangers, Appl. Therm. Eng. 26 (17–18) (2006) 2287–2300.
- [55] W.M. Kays, A.L. London, Compact Heat Exchangers, 3rd ed, Krieger Pub. Co, 1998.
- [56] R.K. Shah, Plate heat exchangers and their design theory. Heat Transfer Equipment Design, CRC Press, 1988.
- [57] J.P. Holman, Experimental Methods For Engineers, 7th ed, McGraw-Hill, 2001.
- [58] B.H. Shon, C.W. Jung, O.J. Kwon, C.K. Choi, Y.Tae Kang, Characteristics on condensation heat transfer and pressure drop for a low GWP refrigerant in brazed plate heat exchanger, Int. J. Heat Mass Transf. 122 (2018) 1272–1282.
- [59] C. Jiang, B. Bai, Flow patterns and pressure drop of downward two-phase flow in a capsule-type plate heat exchanger, Exp. Therm. Fluid Sci. 103 (2019) 347–354.

- [60] S.M. Ghiaasiaan, Two-Phase Flow, Boiling, and Condensation: In Conventional and Miniature Systems, Cambridge University Press, 2007.
- [61] D. Lee, D. Kim, W. Cho, Y. Kim, Evaporation heat transfer and pressure drop characteristics of R1234ze(E)/R32 as a function of composition ratio in a brazed plate heat exchanger, *Int. J. Heat Mass Transf.* 140 (2019) 216–226.
- [62] M. Ciofalo, I. Di Piazza, J. Stasiek, Investigation of flow and heat transfer in corrugated-undulated plate heat exchangers, *Heat Mass Transf.* 36 (5) (2000) 449–462.
- [63] S. Mohebbi, F. Veysi, Numerical investigation of small plate heat exchangers performance having different surface profiles, *Appl. Therm. Eng.* 188 (2021) 116616.
- [64] M. Rossato, D. Del Col, A. Muzzolon, L. Rossetto, Flow boiling of R32 inside a brazed plate heat exchanger, *Int. J. Refrig.* 69 (2016) 165–174.
- [65] R.L. Amalfi, F. Vakili-Farahani, J.R. Thome, Flow boiling and frictional pressure gradients in plate heat exchangers. Part 2: comparison of literature methods to database and new prediction methods, *Int. J. Refrig.* 61 (2016) 185–203.
- [66] J. Ham, Y. Shin, H. Cho, Theoretical investigation of the influence of pipe diameter and exit channel width in welded plate heat exchanger on heat exchanger performance, *Heat Mass Transf.* 56 (2020) 759–771.
- [67] M. Shirzad, M.A. Delavar, S.S.M. Ajarostaghi, K. Sedighi, Evaluation the effects of geometrical parameters on the performance of pillow plate heat exchanger, *Chem. Eng. Res. Des.* 150 (2019) 74–83.